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Liquid Neural Networks (LNNs)

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ဒီနေ့ Talk မှာ သုံးမယ့် အတိုကောက် စာလုံးတွေ

1. LNNs (Liquid Neural Networks)
2. ODE (Ordinary Differential Equations)
3. LTC (Liquid Time Constant)
4. NCP (Neural Circuit Policies)
5. CfC (Closed-form Continuous-time)
6. LSTM တို့ CNN စတဲ့ deep learning နဲ့ ပတ်သက်တာတွေကတော့ သိပြီးသားလို့ ယူဆမယ်

A Simple Analogy of AI

- The function of AI can be compared to solving equations, as its purpose is to resolve complex problems. In cases with two variables and two equations, the process is straightforward.

Example:

$$\text{Equation 1: } x + y = 10$$

$$\text{Equation 2: } x - y = 2$$

A Simple Analogy of AI

- In your school algebra example ($x + 2 = 5$), x is the unknown. In AI, we have parameters (often called **weights** and **biases**).
- Imagine a simple AI trying to predict the price of a house. The "equation" might look like this:

$$\text{Price} = (w_1 \times \text{Square Footage}) + (w_2 \times \text{Number of Bedrooms}) + b$$

- The **Square Footage** and **Bedrooms** are the data you provide.
- The **w1**, **w2**, and **b** are the "unknowns" the AI needs to figure out.

A Simple Analogy of AI

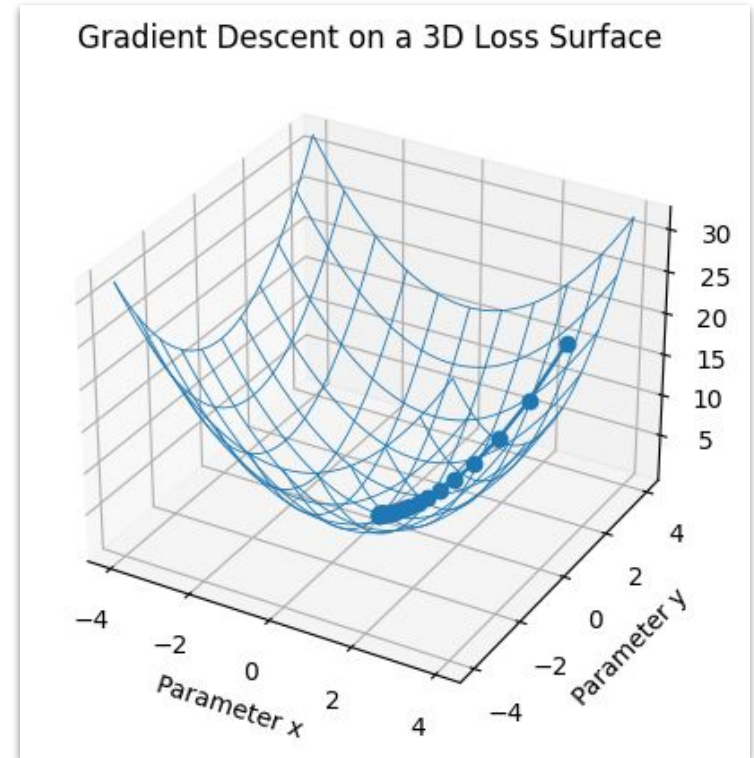
- In algebra, you need at least as many unique equations as you have unknowns to find a single solution. **In AI, every piece of data (like a photo of a cat or a sentence in a book) acts like a constraint or an "equation."**
- **The Goal:** The AI looks at millions of examples and tries to find values for its "unknowns" (weights) that make the equation work for as many examples as possible.
- **The Difference:** Unlike school math, where there is usually one "perfect" answer, AI looks for the best possible fit that minimizes errors across all the data.

A Simple Analogy of AI

- While our example had two unknowns, a model like GPT-4 has trillions of parameters.
- Instead of a simple line on a 2D graph, the AI is trying to find a "shape" in a space with billions of dimensions.
- It isn't just solving for "price"; it's solving for the probability of the next word in a sentence, the texture of fur in an image, or the tone of a voice.
- We also called “**Optimization**”.

A Simple Analogy of AI

- How the "Solving" Actually Happens?
- We use a process called **Gradient Descent**.
- Start with a Guess, Check the error (i.e. Loss Function), Adjust the weight, Repeat (billions of times until the equation is accurate as it can get)



Liquid Neural Networks (LNNs)

- Building on the "equation solving" analogy, Liquid Neural Networks (LNNs) are the next evolution. If traditional AI is like solving a static equation, **Liquid AI is like solving a living equation.**
- Traditional AI as a "line of best fit" on a graph. Once the AI is trained, that line is frozen.
- **Traditional AI (Solid):** Think of it like a concrete sculpture. It's incredibly detailed, but if you want to change its shape to fit a new room, you have to smash it and build a new one (retraining).
- **Liquid AI (Fluid):** Think of it like a pool of water or a piece of clay. As new information (input) flows in, the "equation" itself ripples and adjusts its shape in real-time. It doesn't just solve the equation once; it keeps solving it as the variables change.

Liquid Neural Networks (LNNs)

- Liquid Neural Networks use Ordinary Differential Equations (ODEs). Instead of the answer being a fixed number, the answer is a description of how things change over time.
- **The Benefit:** Because the math is designed to handle "change," the AI can deal with unexpected events (like a sudden rainstorm while driving or a glitch in a sensor) without getting confused. It just "flows" with the new data.

Liquid Neural Networks (LNNs)

- One of the most mind-blowing facts about LNNs (mentioned in the MIT research by Ramin Hasani and Daniela Rus) is their size.
- **Traditional AI:** A self-driving car model might need 100,000 artificial neurons to stay in its lane.
- **Liquid AI:** Researchers created a Liquid Network that could steer a car using **only 19 neurons**.
- **How is that possible?** They looked at **a tiny worm called C. elegans**. It has only 302 neurons, yet it can navigate, find food, and survive in a complex world.

Liquid Neural Networks (LNNs)

- Summary: The “Equation” Comparison

Feature	Traditional AI	Liquid Neural Networks
Math Style	Static Algebra (Fixed weights)	Dynamic Calculus (ODEs)
After Training	Frozen / Fixed	Continues to adapt
Size	Massive (Millions/Billions of nodes)	Tiny (Dozens/Hundreds of nodes)
Best For	Stable data (Text, static images)	Moving data (Video, sensors, robots)

Liquid Neural Networks (LNNs)

- Traditional RNN: the hidden state h at step t is a simple jump:

$$h_t = \sigma(Wx_t + Uh_{t-1} + b)$$

- x_t : The current word (e.g., "ကျော့").
- h_{t-1} : The previous memory (e.g., "ဆူ").
- The Problem:** The weights W and U are fixed. The "internal clock" is fixed to the word count.

Liquid Neural Networks (LNNs)

- The hidden state is the solution to a Differential Equation:

$$\frac{dh(t)}{dt} = - \underbrace{\left[\frac{1}{\tau} + f(x(t), \theta) \right]}_{\text{"Liquid" Time Constant}} h(t) + \underbrace{f(x(t), \theta) A}_{\text{Input Target}}$$

- $1/\tau$: The "leakage." How fast the neuron forgets.
- $f(x, \theta)$: This is the "Liquid" part. The input word ("ᄠᆞᆫᆫᆞᆫ") actually changes the speed of the neuron's reaction.
- **Continuous Time:** We can calculate $h(t)$ at any moment, even halfway between words.

Liquid Neural Networks (LNNs)

- Visualizing the "invisible math" happening inside two different networks



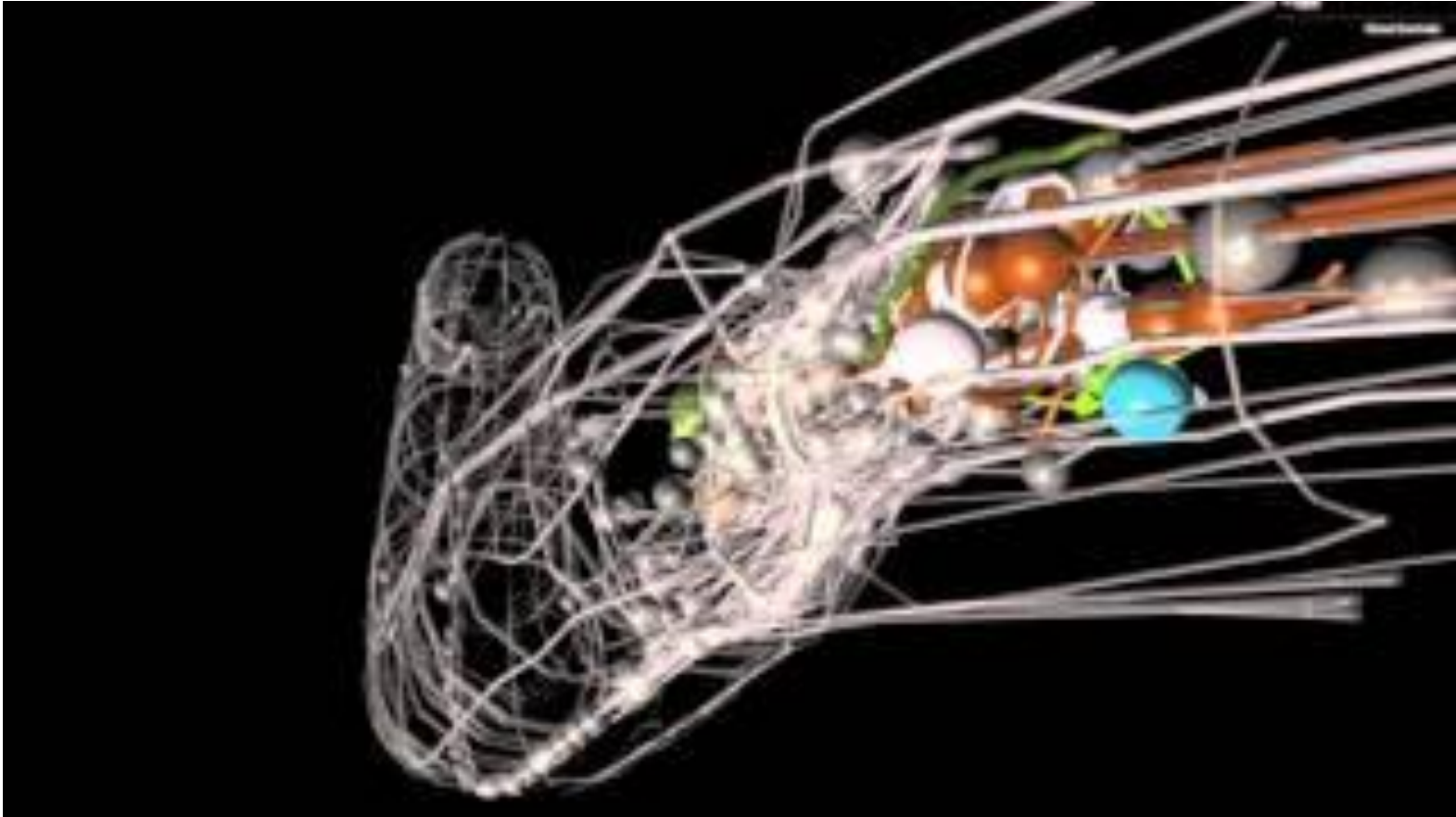
Sources: Wikipedia

Liquid Neural Networks (LNNs)

Concept	Digital Clock (Box)	Hourglass (Liquid)
Logic	"If it's Time 1, do Task A. If it's Time 2, do Task B."	"How much has changed since the last time I looked?"
Math	Algebra ($y=mx+b$)	Calculus ($dt/dy=\text{flow}$)

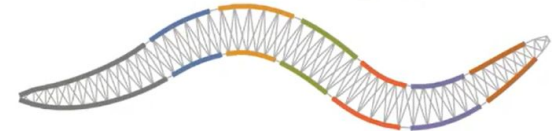
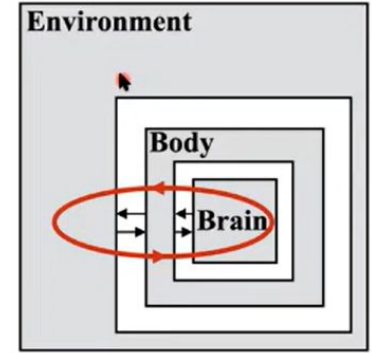
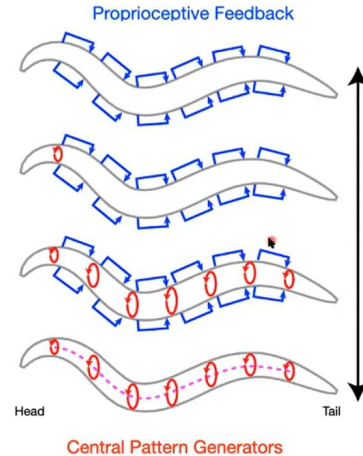
- In our analogy, the **Hourglass is the Neuron**,
- the **Sand is the information** (the state)
- the **Weather is the Input Data**.

The Evolution of LNNs: From Worms to ODEs



The Evolution of LNNs: From Worms to ODEs

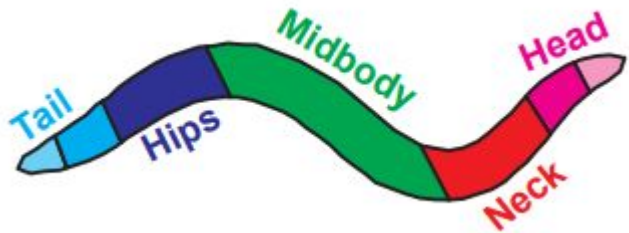
- Microscopic roundworm (nematode), ~1 mm long
- Found in temperate soil environments worldwide
- Feeds on bacteria; in labs, fed *E. coli*
- Transparent body enables direct observation of cells



Sources: <https://www.youtube.com/watch?v=IXrtoqu4Uso>

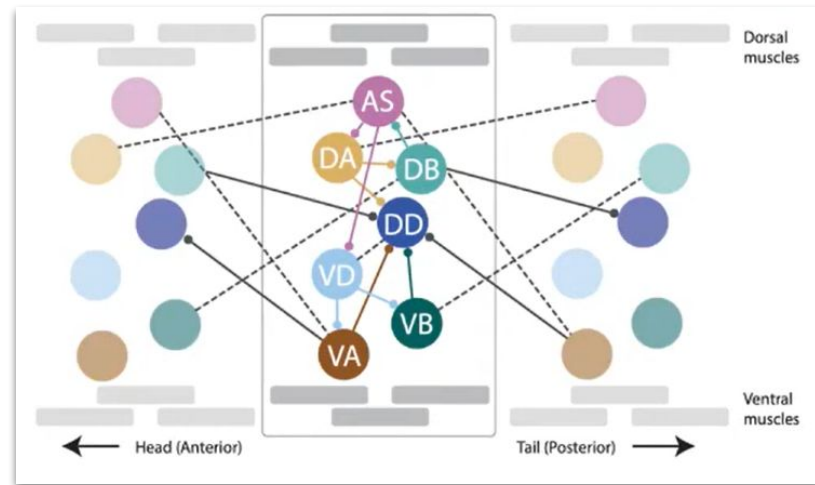
The Evolution of LNNs: From Worms to ODEs

- Two forms: hermaphrodite (959 cells) and male (1,031 cells)
- Nervous system contains only 302 neurons
- Entire neural connectome fully mapped
- Extremely simple compared to the human brain



Sources:

https://www2.mrc-lmb.cam.ac.uk/groups/wschafer/wp-content/uploads/sites/38/tjucikas_phd_thesis.pdf



Sources: <https://www.youtube.com/watch?v=IXrtoqu4Uso>

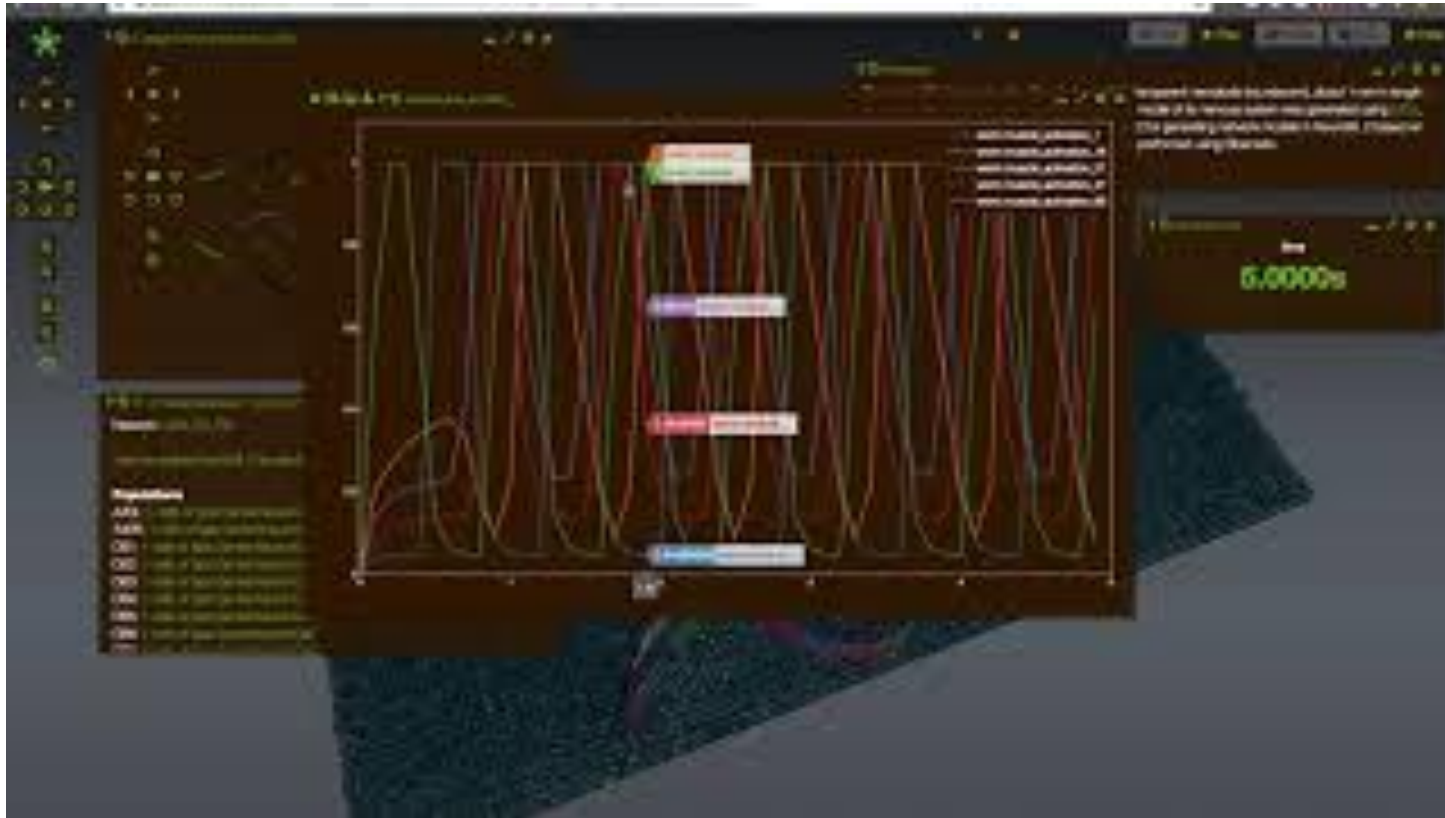
The Evolution of LNNs: From Worms to ODEs

- Primitive behaviors: feeding, locomotion, reproduction
- Complex behaviors: learning, mating, social interaction
- Behavior reflects nervous system activity
- Powerful model organism for studying biology and disease



Sources: <https://takeishilab.com/>

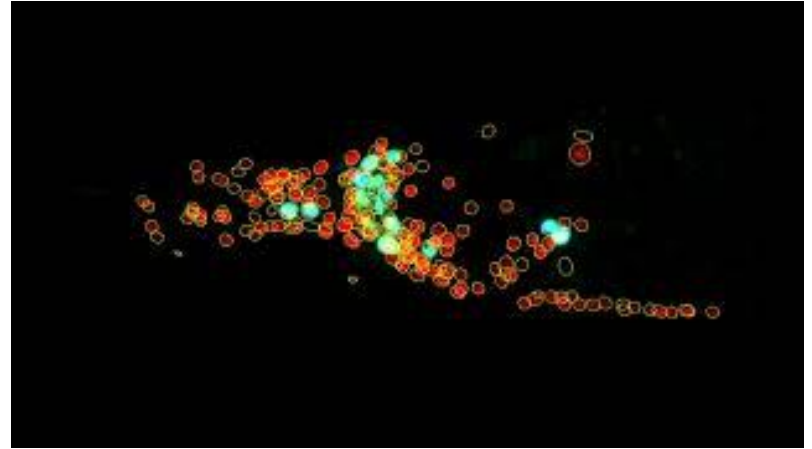
The Evolution of LNNs: From Worms to ODEs



Sources: <https://www.youtube.com/watch?v=M0rpT831ngM>

The Evolution of LNNs: From Worms to ODEs

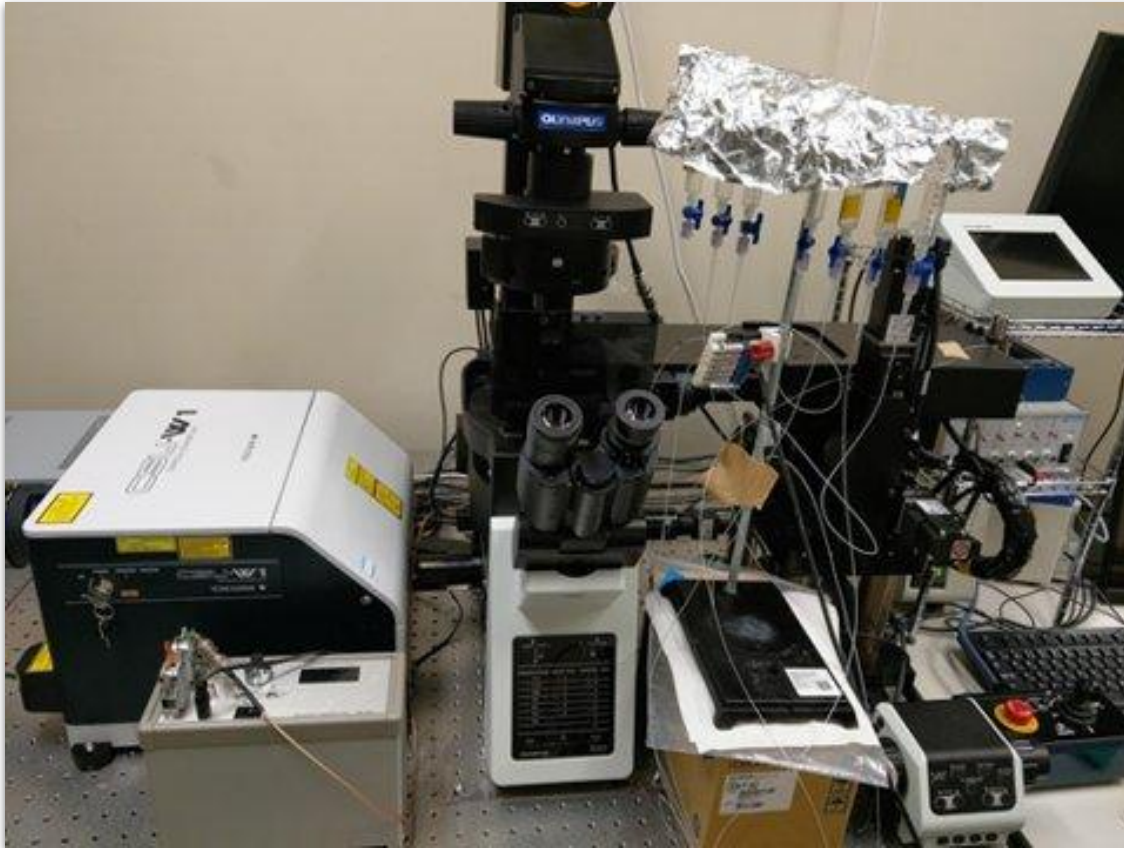
- Whole-brain, single-cell–resolution imaging is key to understanding information processing
- *C. elegans* is ideal due to its small, transparent body
- Neural activity is recorded in living worms using fluorescence microscopy and microfluidics
- A 4D microscope enables simultaneous observation of all head neurons
- Image analysis and cell identification map neural activity to known circuits



Sources:

<https://www.youtube.com/watch?v=ycJ-vcBdEXU&t=19s>

The Evolution of LNNs: From Worms to ODEs

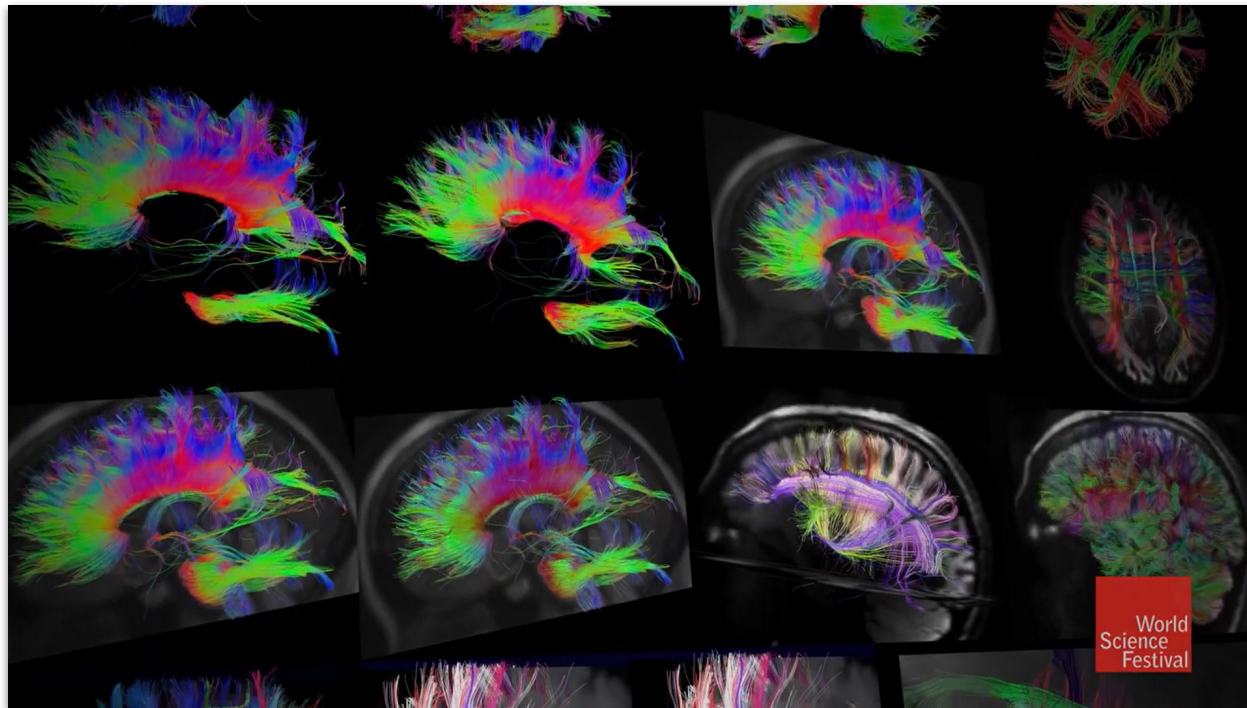


- 4D Microscope
(i.e 3D space +
time)

Sources:

<https://www.bs.s.u-tokyo.ac.jp/~toyoshimalab/en/research/index.html>

The Evolution of LNNs: From Worms to ODEs



- Structural connections
- Functional connections
- a 1 millimeter cube of brain tissue contains 50,000 nerve cells and about 1 billion synapses
- Currently, no way to model an entire human brain by diagramming its micro wiring

The Evolution of LNNs: From Worms to ODEs



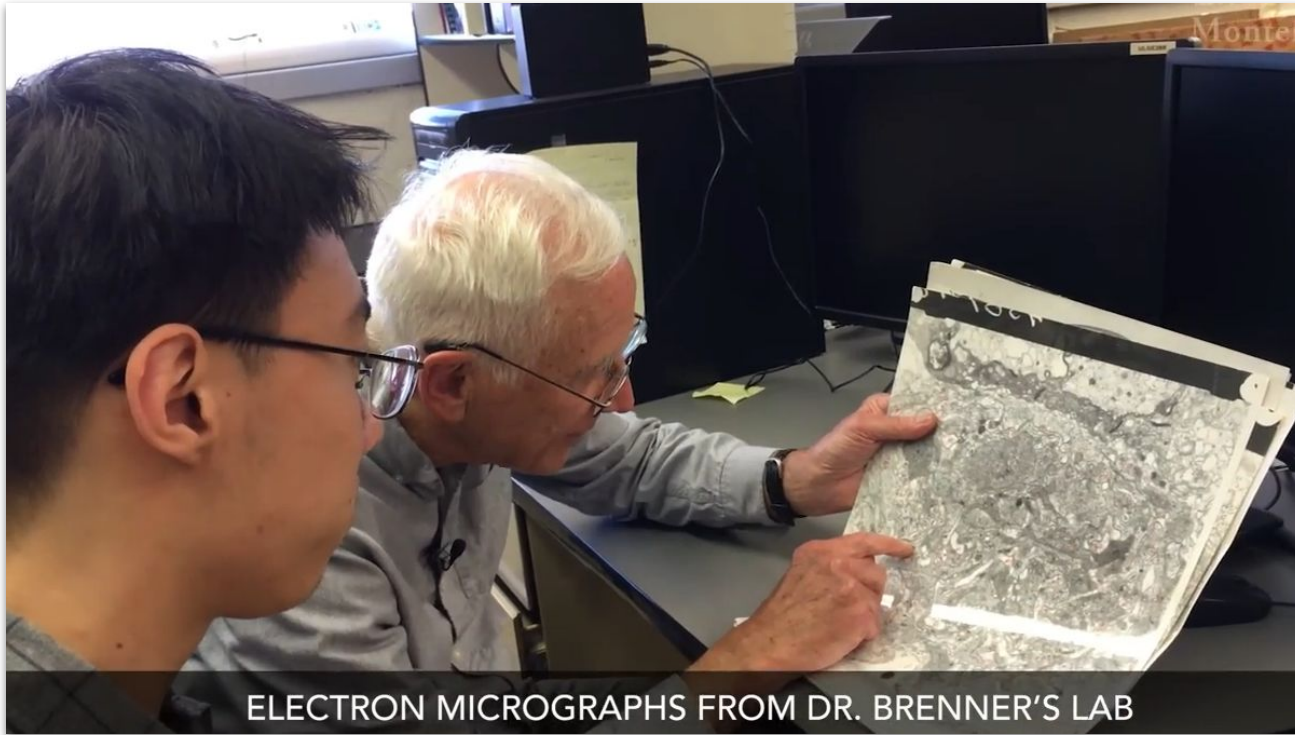
Volume 571 Issue 7763, 4 July 2019

- Knowing how the components of a nervous system are connected together is crucial for understanding how that system works. In this week's issue, **Scott Emmons and his colleagues present complete wiring diagrams** for both sexes of the model organism *Caenorhabditis elegans*. These neural maps, or connectomes, which update an influential 1986 publication, were built up from both new and previously published electron micrographs and encompass all connections in the nematode from sensory input to end-organ output. The maps have allowed the researchers to determine the location of each synapse and to assign an indirect measure of the strength of each connection based on the physical size of its constituent synapses.

Sources:

<https://www.nature.com/nature/volumes/571/issues/7763>

The Evolution of LNNs: From Worms to ODEs



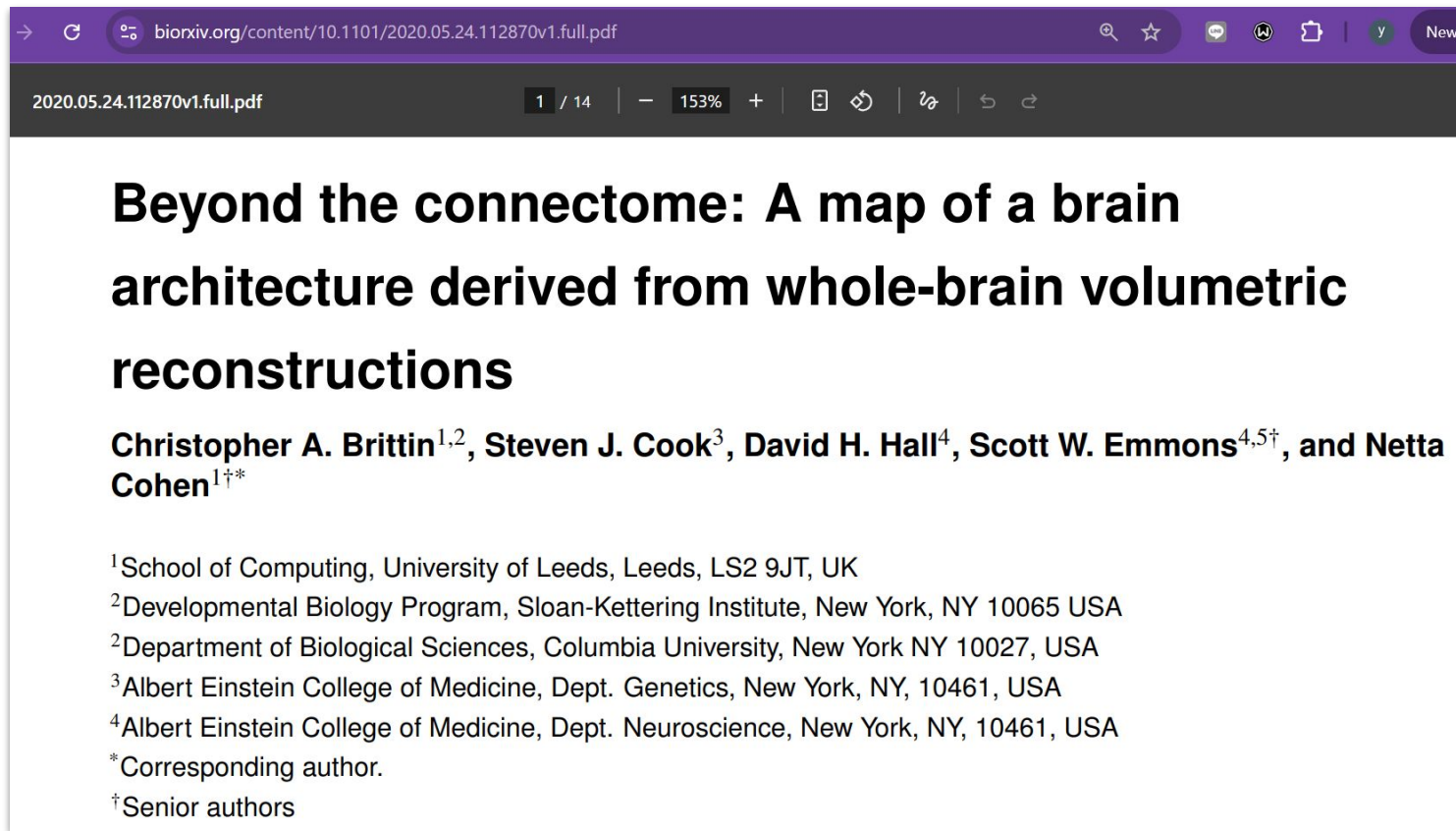
- The role of *C. Elegans*

Sources:

<https://www.youtube.com/watch?v=LBKEyGaPamY>

- Dr. Scott Emmons and colleagues continued analyzing

The Evolution of LNNs: From Worms to ODEs



The image is a screenshot of a web browser displaying a PDF document from bioRxiv. The address bar shows the URL: <https://www.biorxiv.org/content/10.1101/2020.05.24.112870v1.full.pdf>. The PDF viewer interface includes a toolbar with navigation and zoom controls. The document title is "Beyond the connectome: A map of a brain architecture derived from whole-brain volumetric reconstructions". The authors listed are Christopher A. Brittin^{1,2}, Steven J. Cook³, David H. Hall⁴, Scott W. Emmons^{4,5†}, and Netta Cohen^{1†*}. Below the authors, their affiliations are provided: ¹School of Computing, University of Leeds, Leeds, LS2 9JT, UK; ²Developmental Biology Program, Sloan-Kettering Institute, New York, NY 10065 USA; ²Department of Biological Sciences, Columbia University, New York NY 10027, USA; ³Albert Einstein College of Medicine, Dept. Genetics, New York, NY, 10461, USA; ⁴Albert Einstein College of Medicine, Dept. Neuroscience, New York, NY, 10461, USA. Footnotes indicate that * is the corresponding author and † are senior authors.

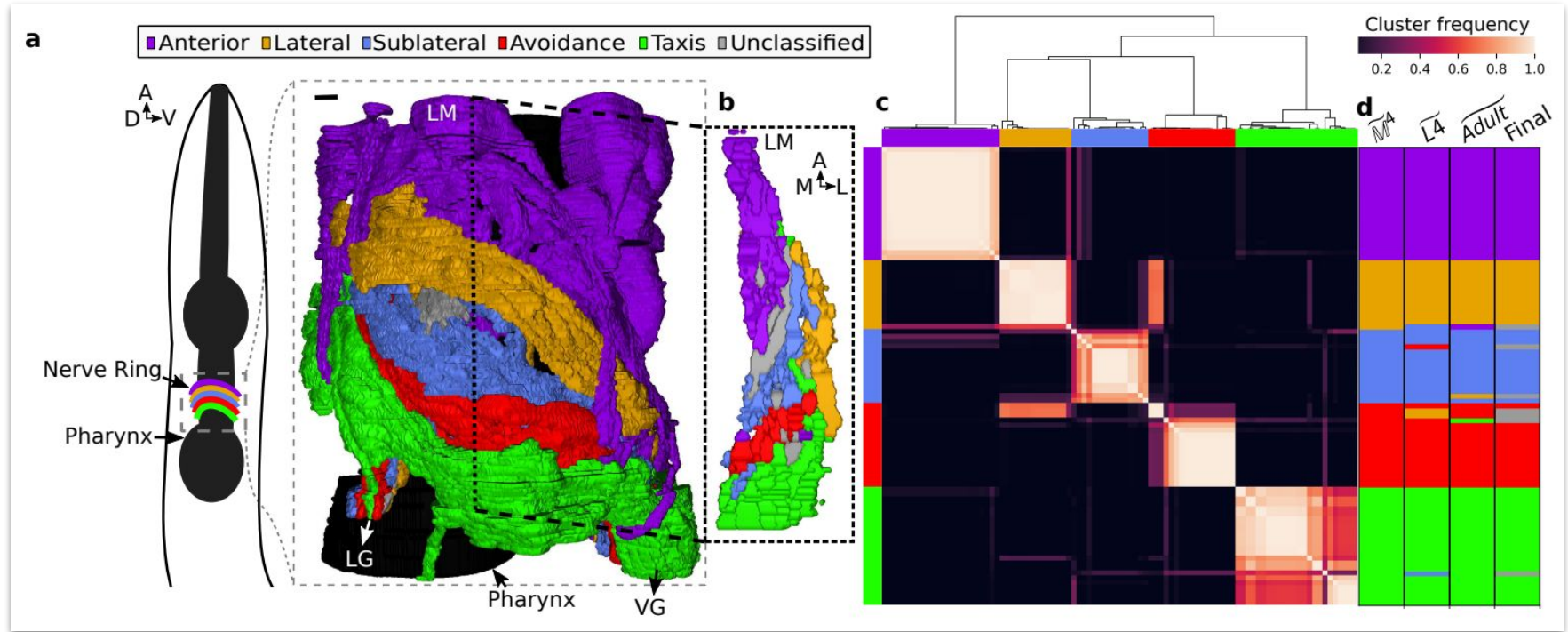
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Beyond the connectome: A map of a brain architecture derived from whole-brain volumetric reconstructions

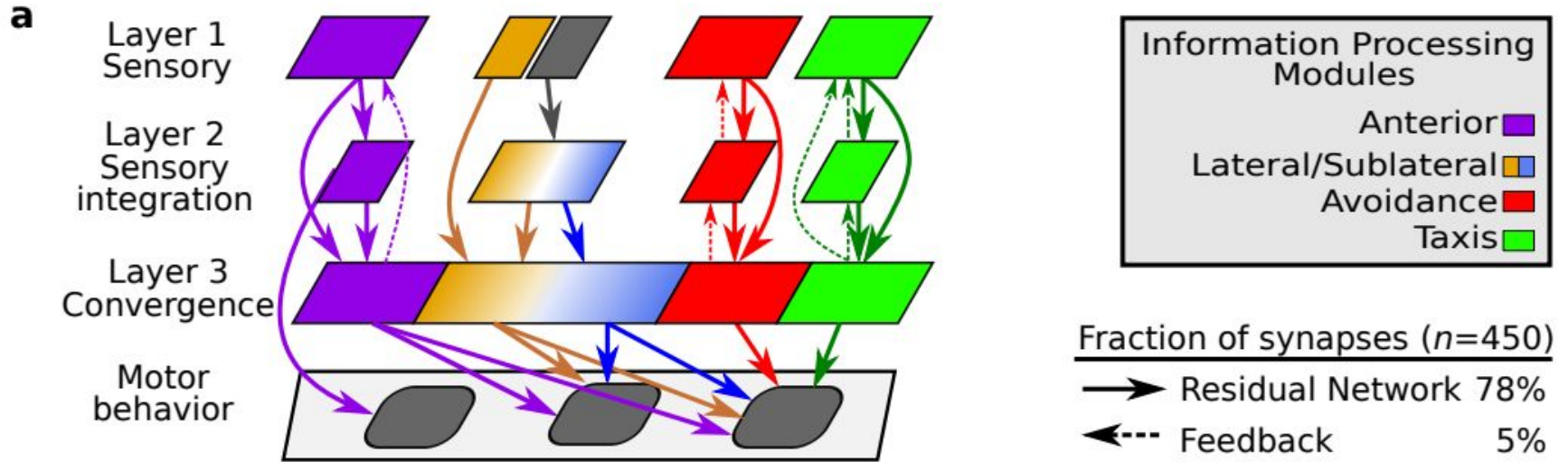
Christopher A. Brittin^{1,2}, Steven J. Cook³, David H. Hall⁴, Scott W. Emmons^{4,5†}, and Netta Cohen^{1†*}

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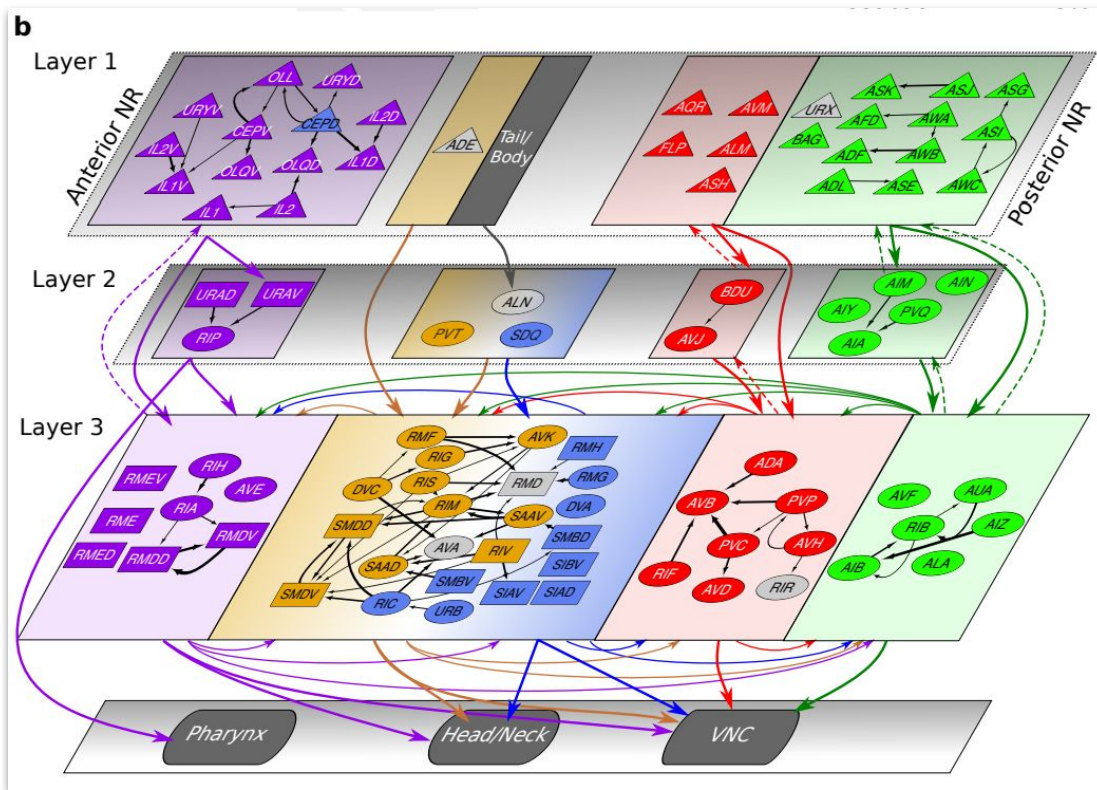
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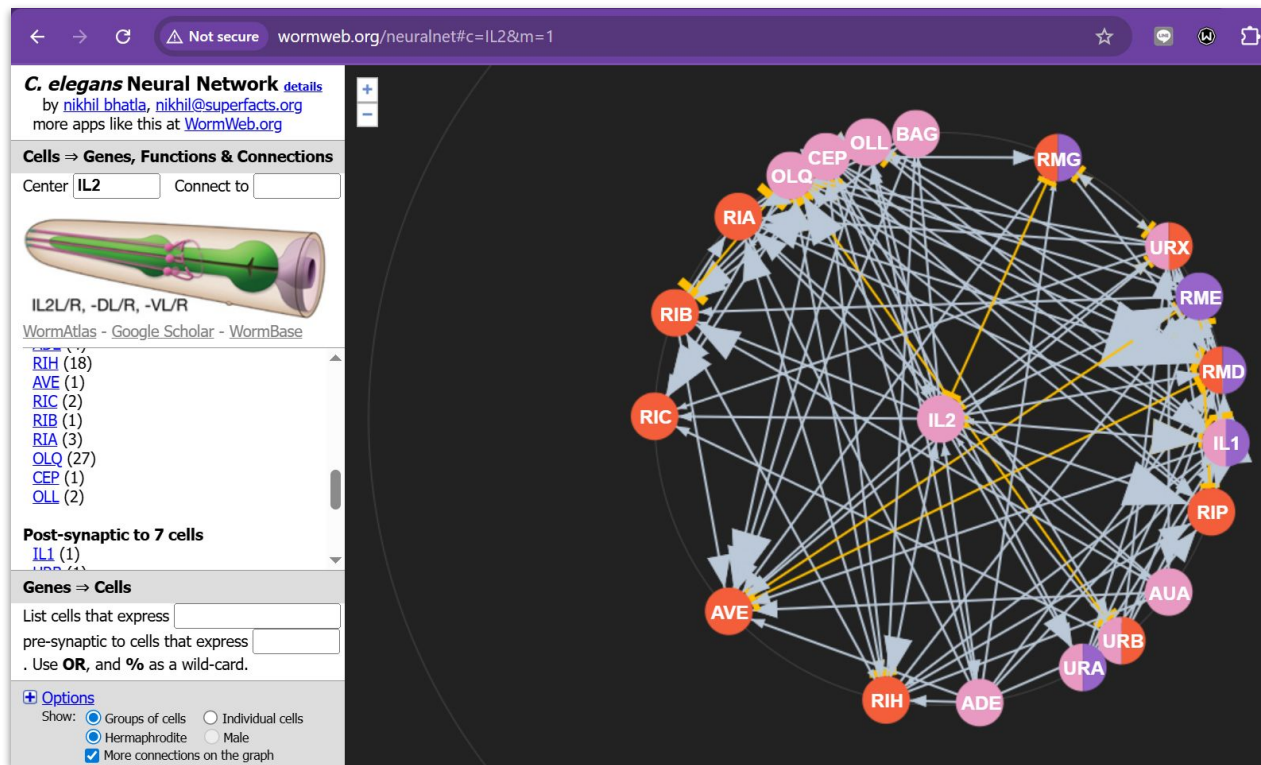


The Evolution of LNNs: From Worms to ODEs



- Every single neuron has a unique 2-to-4-letter name
- For example, RMEV: Ring Motor Neuron, Efferent, Ventral, RIH: Ring Interneuron, Head
- Triangles: sensory neurons that receive input from the environment (touch, chemicals, light)
- Ovals: interneurons (processing units)
- Rectangles: motoneurons

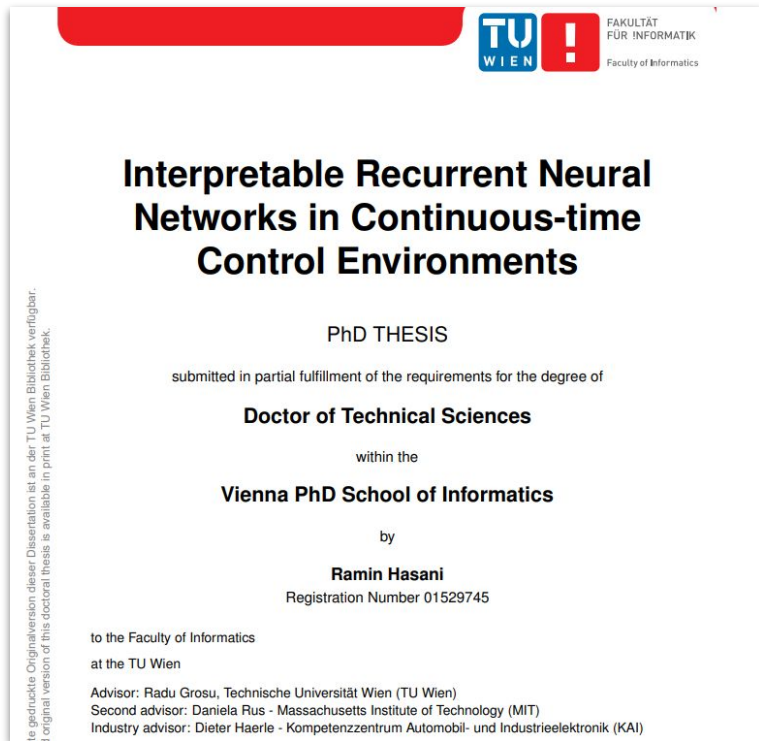
The Evolution of LNNs: From Worms to ODEs



- **PVQ:** Posterior Ventral Q-cell (Interneuron); handles sensory integration and sexual circuits.
- **AVF:** Anterior Ventral F-cell (Interneuron); involved in coordinating locomotion and egg-laying.

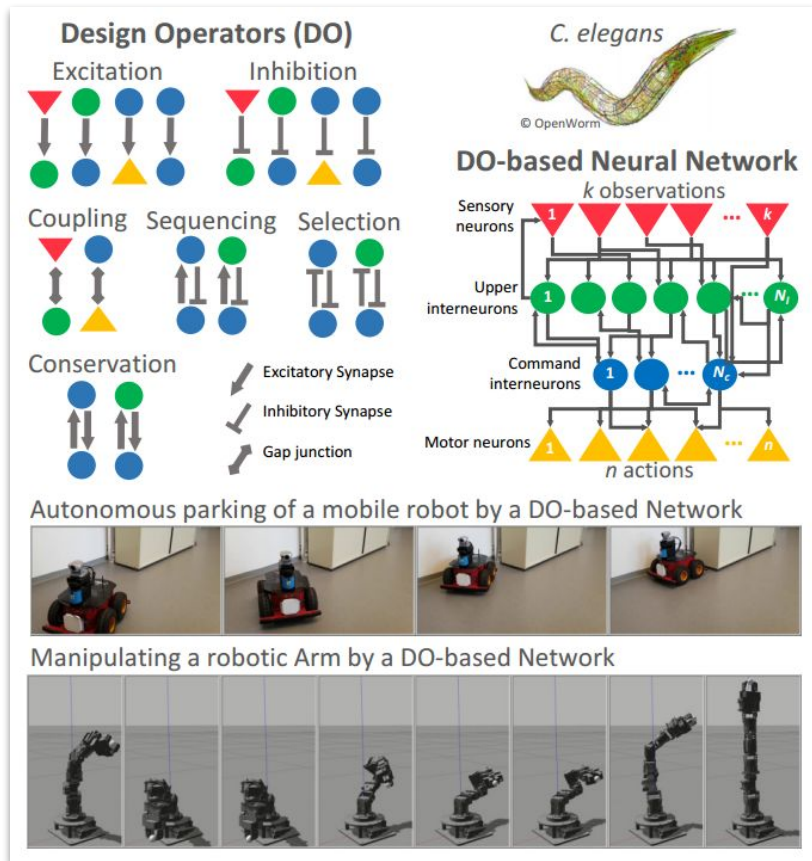
Sources: <http://wormweb.org/neuralnet#c=IL2&m=1>

The Evolution of LNNs: From Worms to ODEs



- We invented liquid neural networks, a class of brain-inspired systems that can stay adaptable and robust to changes even after training [R. Hasani, PhD Thesis] [Lechner et al. Nature MI, 2020] [pdf] (2016-2020).
- We then analytically and experimentally showed they are universal approximators [Hasani et al. AAAI, 2021], expressive continuous-time machine learning systems for sequential data [Hasani et al. AAAI, 2021] [Hasani et al. Nature MI, 2022], parameter efficient in learning new skills [Lechner et al. Nature MI, 2020] [pdf], causal and interpretable [Vorbach et al. NeurIPS, 2021] [Chahine et al. Science Robotics 2023] [pdf], and when linearized they can efficiently model very long-term dependencies in sequential data [Hasani et al. ICLR 2023].

The Evolution of LNNs: From Worms to ODEs



- We design **compact, interpretable and noise-robust** neural controllers inspired by the relational structures of the worm's brain, to control robots
- Experimental demonstrations of the superiority of the performance of DO-based networks in terms of their compactness, robustness to noise, and their interpretable dynamics, compared to contemporary RNN architectures.

The Evolution of LNNs: From Worms to ODEs

Liquid Time-Constant Networks

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Abstract

We introduce a new class of time-continuous recurrent neural network models. Instead of declaring a learning system's dynamics by implicit nonlinearities, we construct networks of linear first-order dynamical systems modulated via nonlinear interlinked gates. The resulting models represent dynamical systems with varying (i.e., *liquid*) time-constants coupled to their hidden state, with outputs being computed by numerical differential equation solvers. These neural networks exhibit stable and bounded behavior, yield superior expressivity within the family of neural ordinary differential equations,

by the following equation (Funahashi and Nakamura 1993): $\frac{d\mathbf{x}(t)}{dt} = -\frac{\mathbf{x}(t)}{\tau} + f(\mathbf{x}(t), \mathbf{I}(t), t, \theta)$, in which the term $-\frac{\mathbf{x}(t)}{\tau}$ assists the autonomous system to reach an equilibrium state with a time-constant τ . $\mathbf{x}(t)$ is the hidden state, $\mathbf{I}(t)$ is the input, t represents time, and f is parametrized by θ .

We propose an alternative formulation: let the hidden state flow of a network be declared by a system of linear ODEs of the form: $d\mathbf{x}(t)/dt = -\mathbf{x}(t)/\tau + \mathbf{S}(t)$, and let $\mathbf{S}(t) \in \mathbb{R}^M$ represent the following nonlinearity determined by $\mathbf{S}(t) = f(\mathbf{x}(t), \mathbf{I}(t), t, \theta)(A - \mathbf{x}(t))$, with parameters θ and A . Then, by plugging in $\mathbf{S}$ into the hidden states equation we get:

The Evolution of LNNs: From Worms to ODEs

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Article | [Open access](#) | Published: 15 November 2022

Closed-form continuous-time neural networks

[Ramin Hasani](#) , [Mathias Lechner](#), [Alexander Amini](#), [Lucas Liebenwein](#), [Aaron Ray](#), [Max Tschaikowski](#), [Gerald Teschl](#) & [Daniela Rus](#)

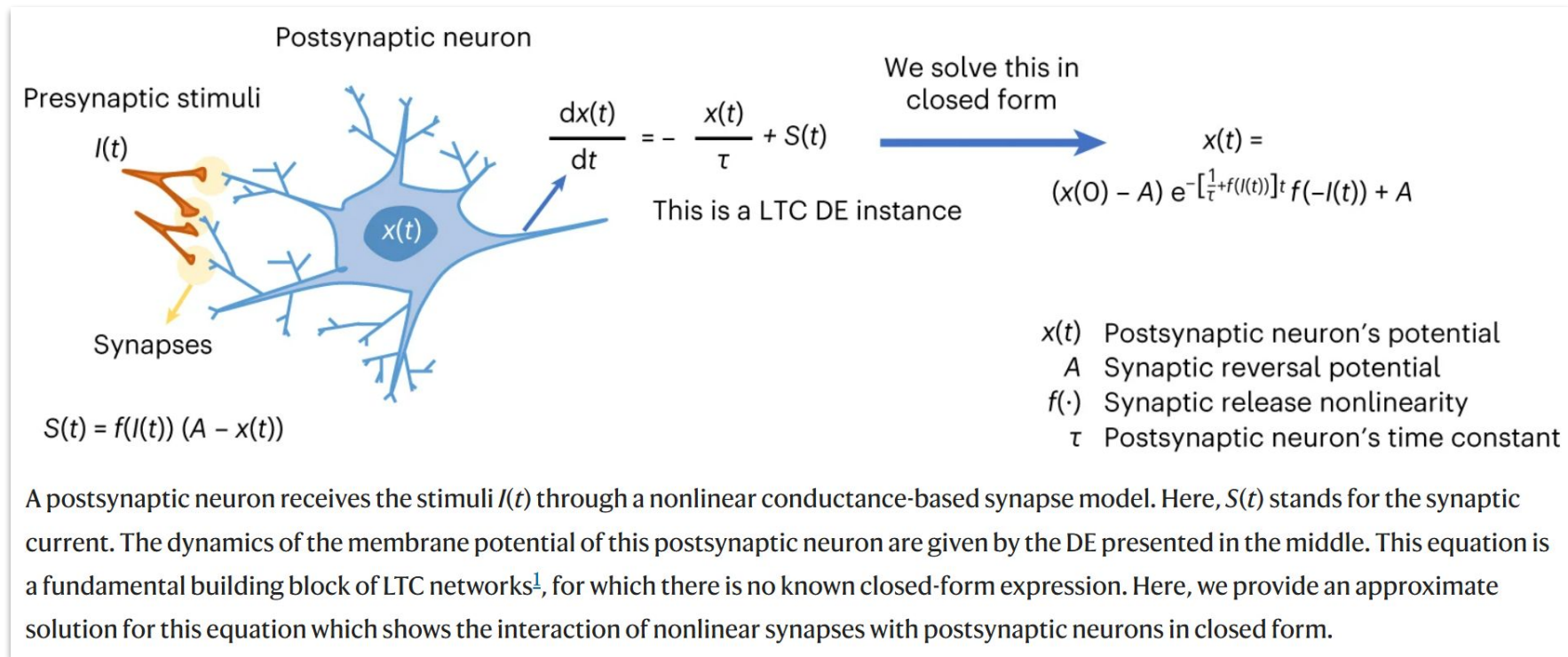
[Nature Machine Intelligence](#) **4**, 992–1003 (2022) | [Cite this article](#)

157k Accesses | 127 Citations | 252 Altmetric | [Metrics](#)

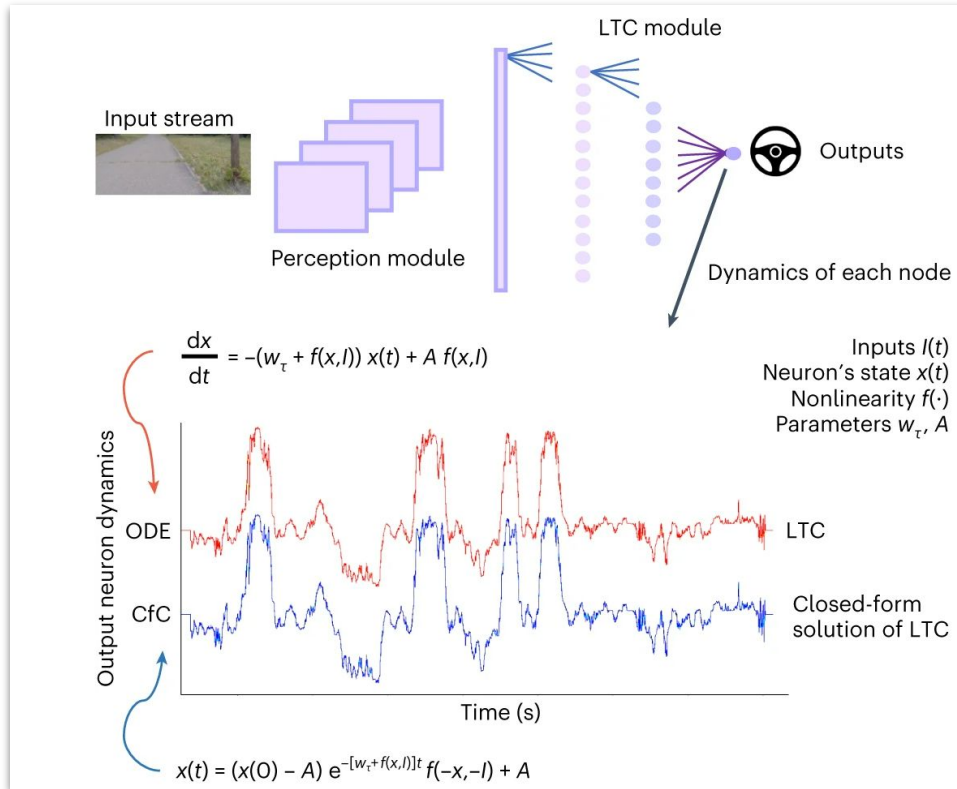
 A [Publisher Correction](#) to this article was published on 29 November 2022

- နောက်ပိုင်း LNNs
မော်ဒယ်တွေအားလုံး
က ဒီ CfC
နည်းလမ်းကိုပဲ
အခြေခံတယ်

The Evolution of LNNs: From Worms to ODEs

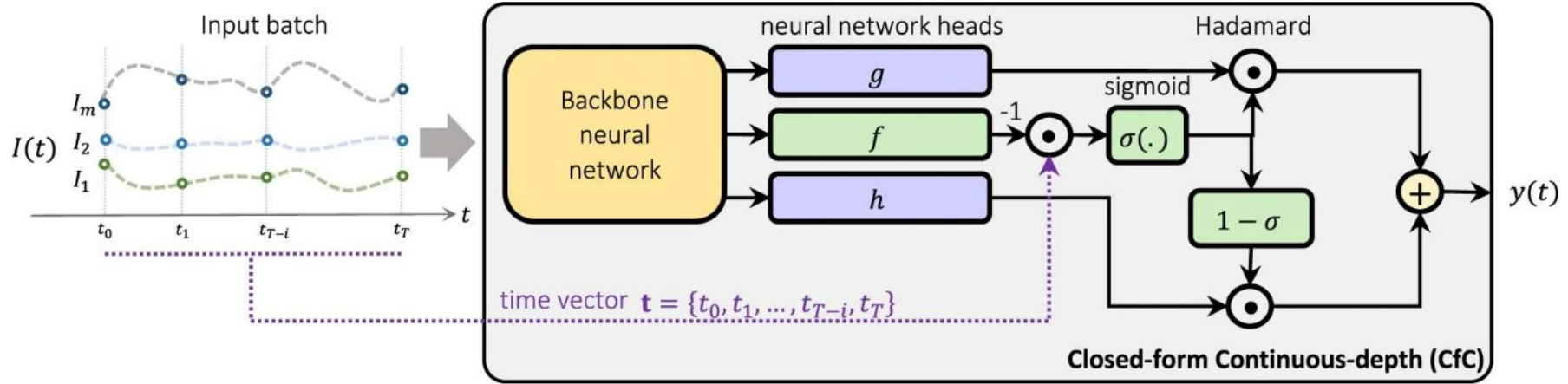


The Evolution of LNNs: From Worms to ODEs



- ဒီ graph ရလဒ် ကနေပြီး
CfC နဲ့ ODE ရလဒ်က
ဘယ်လောက်ထိ
နီးစပ်သလဲ ဆိုတာကို
မြင်ရပါလိမ့်မယ်

The Evolution of LNNs: From Worms to ODEs



A backbone neural network layer delivers the input signals into three head networks g , f and h . f acts as a *liquid* time-constant for the sigmoidal time-gates of the network. g and h construct the nonlinearities of the overall CfC network.

Source: <https://www.nature.com/articles/s42256-022-00556-7/figures/4>

- ဒီပုံကတော့ လက်တွေ့ implementation/training လုပ်တဲ့အခါမှာ ဘယ်လိုလုပ်သလဲ ဆိုတာကို ရှင်းပြတဲ့ ပုံပါ။

The Evolution of LNNs: From Worms to ODEs



Ramin Hasani ✓ • Following

Co-founder & CEO @ Liquid AI

2yr • 🌐

[United States Air Force](#) successfully flew an X-62 VISTA jet aircraft (derived from the F-16D Fighting Falcon) autonomously by liquid neural networks!

This is an important step towards achieving explainable, safe, and robust autonomy at scale!

So proud of my co-inventors and teammates [Mathias Lechner](#), [Tsun-Hsuan Wang](#), [Alexander Amini](#), and [Daniela Rus](#) at the [Massachusetts Institute of Technology](#) and [Liquid AI](#)!

Hell of a job by our colleagues at the [MIT Lincoln Laboratory](#), [AI Accelerator](#), [Defense Advanced Research Projects Agency \(DARPA\)](#), and [412th Test Wing, Edwards Air Force Base](#)! Special thanks to the amazing [Josh "Voodoo" Rountree](#) for leading the effort and trusting our technology!

Sources:

[linkedin.com](https://www.linkedin.com)

The Evolution of LNNs: From Worms to ODEs



Sources: [linkedin.com](https://www.linkedin.com)

The Evolution of LNNs: From Worms to ODEs

SCIENCE ROBOTICS | RESEARCH ARTICLE

ARTIFICIAL INTELLIGENCE

Robust flight navigation out of distribution with liquid neural networks

Makram Chahine[†], Ramin Hasani^{†*}, Patrick Kao[†], Aaron Ray[†], Ryan Shubert, Mathias Lechner, Alexander Amini, Daniela Rus

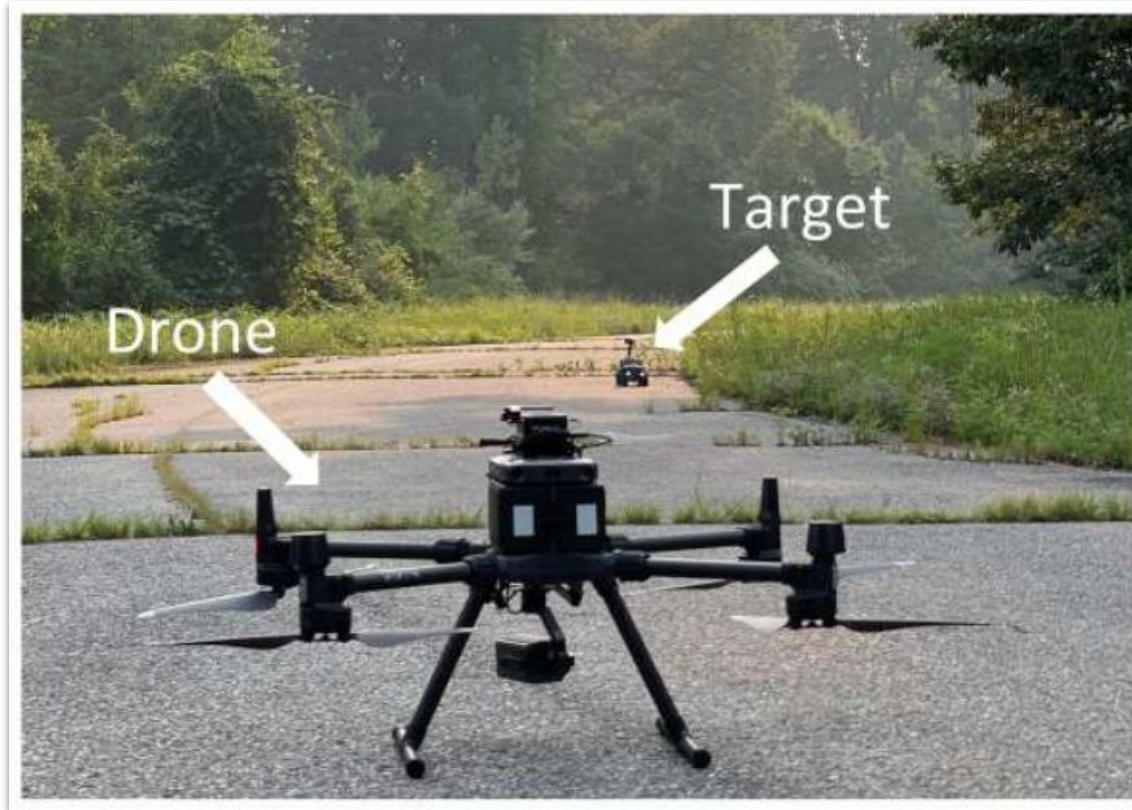
Autonomous robots can learn to perform visual navigation tasks from offline human demonstrations and generalize well to online and unseen scenarios within the same environment they have been trained on. It is challenging for these agents to take a step further and robustly generalize to new environments with drastic scenery changes that they have never encountered. Here, we present a method to create robust flight navigation agents that successfully perform vision-based fly-to-target tasks beyond their training environment under drastic distribution shifts. To this end, we designed an imitation learning framework using liquid neural networks, a brain-inspired class of continuous-time neural models that are causal and adapt to changing conditions. We observed that liquid agents learn to distill the task they are given from visual inputs and drop irrelevant features. Thus, their learned navigation skills transferred to new environments. When compared with several other state-of-the-art deep agents, experiments showed that this level of robustness in decision-making is exclusive to liquid networks, both in their differential equation and closed-form representations.

- OOD ကို လက်တွေ့မှာ မော်ဒယ်တော်တော်များများက fail ဖြစ်

Sources:

<https://cap.csail.mit.edu/sites/default/files/research-pdfs/Robust%20flight%20navigation%20out%20of%20distribution%20with%20liquid%20neural%20networks.pdf>

The Evolution of LNNs: From Worms to ODEs



- A: Task - Visual navigation to target

The Evolution of LNNs: From Worms to ODEs



- B: Test - Robust task understanding under heavy distribution shift

The Evolution of LNNs: From Worms to ODEs



- C: Samples from train set (drone view)

The Evolution of LNNs: From Worms to ODEs



- D: Samples from the four test environments (drone view)

The Evolution of LNNs: From Worms to ODEs

Table 1. Closed-loop evaluation of the fly-to-target task for the trained policies in four testing environments. Quantitative results are success rates. Higher is better. ($N = 40$)

Algorithm	Environment			
	Training Woods	Alt. Woods	Urban Lawn	Urban Patio
LSTM	82.5%	92.5%	62.5%	27.5%
GRU-ODE	50%	17.5%	32.5%	17.5%
ODE-RNN	62.5%	82.5%	17.5%	25%
TCN	7.5%	0%	0%	0%
NCP (ours)	100%	95%	57.5%	52.5%
CfC (ours)	85%	95%	90%	67.5%

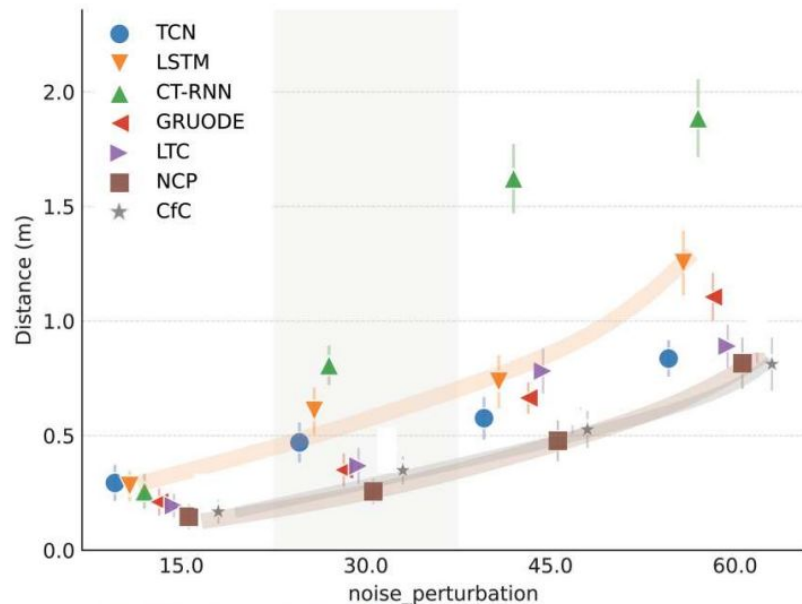
The Evolution of LNNs: From Worms to ODEs

Table 2. Range test evaluation of the trained policies in the Training Woods environment. Quantitative results are success rates. Higher is better. ($N = 10$)

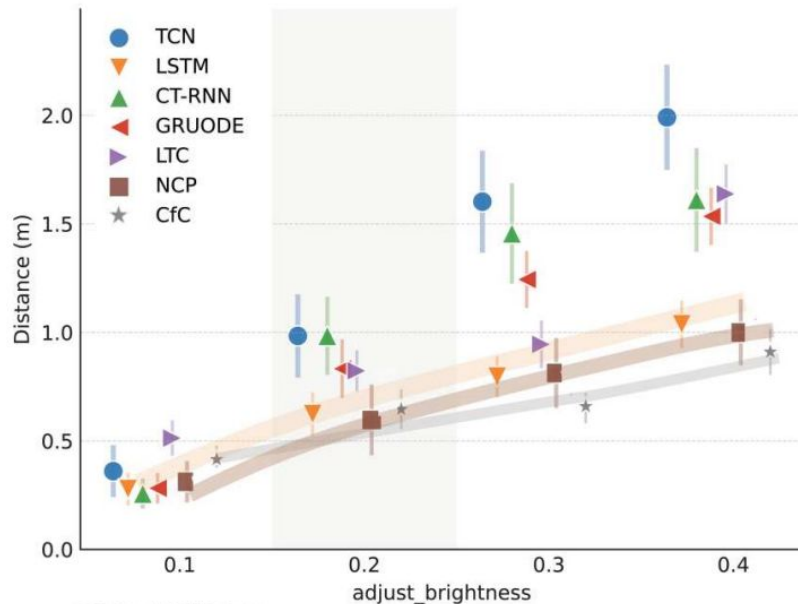
Algorithm	Initial distance (m)		
	10	20	30
LSTM	100%	0%	0%
GRU-ODE	70%	0%	0%
ODE-RNN	80%	10%	0%
TCN	5%	0%	0%
NCP (ours)	100%	50%	10%
CfC (ours)	100%	90%	20%

The Evolution of LNNs: From Worms to ODEs

A Noise perturbation

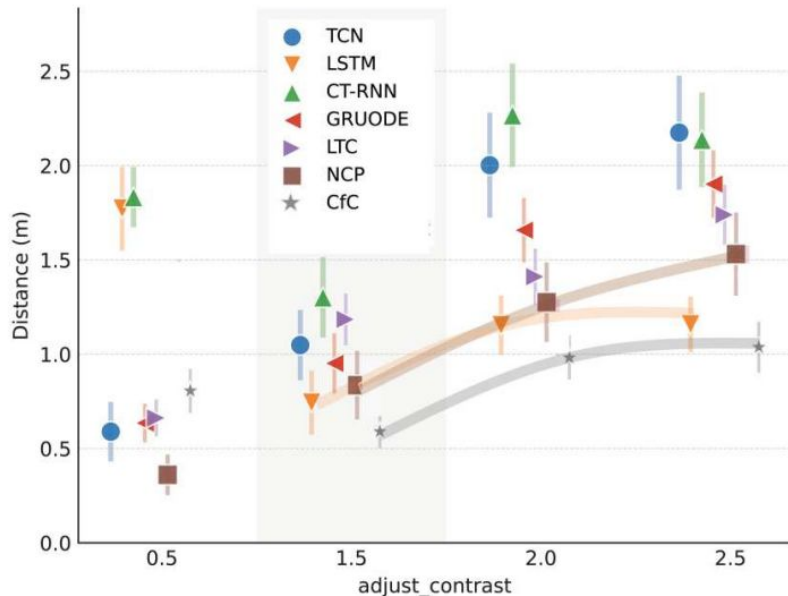


B Robustness to Brightness

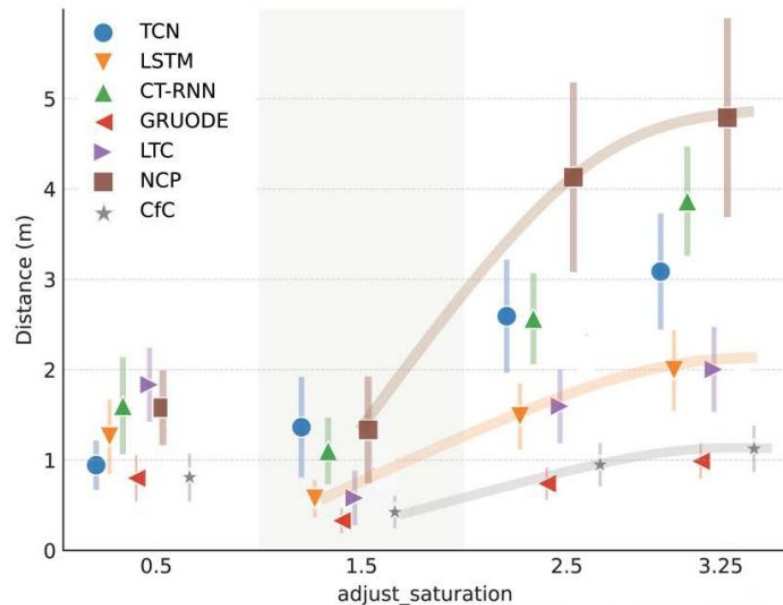


The Evolution of LNNs: From Worms to ODEs

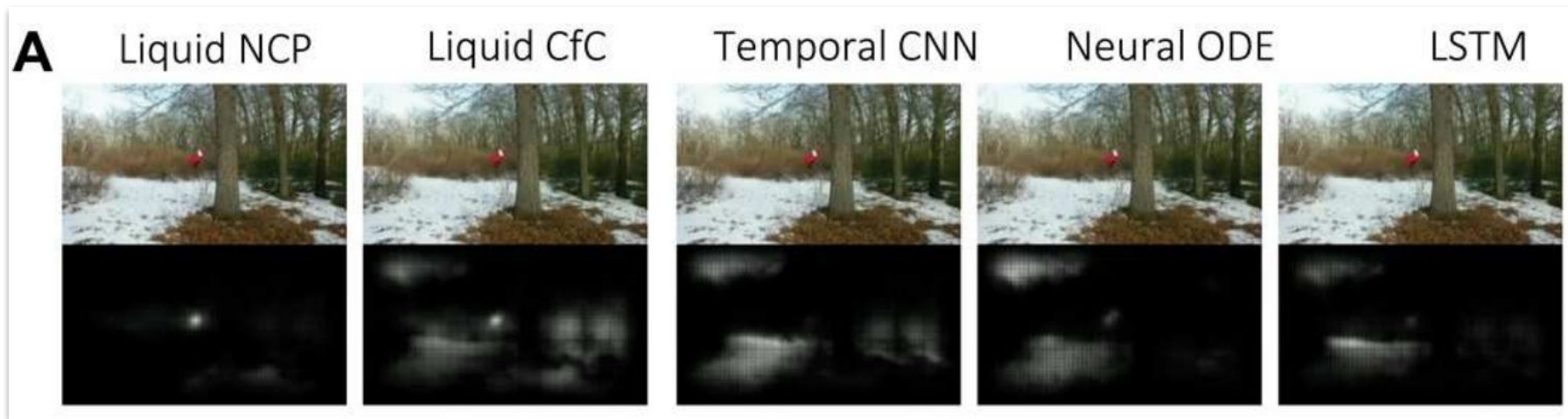
C Robustness to Contrast



D Robustness to Saturation



The Evolution of LNNs: From Worms to ODEs



- A frame illustration of the attention profile of different networks computed by the VisualBackProp saliency map computation algorithm. The top row shows the input image frame. The bottom row shows the saliency map. The brighter regions indicate a greater concentration of the convolutional layer's attention. Saliency scores range between 0 (black) and 1 (white).

The Evolution of LNNs: From Worms to ODEs

B LSTM loses the target and get confused by an adversary



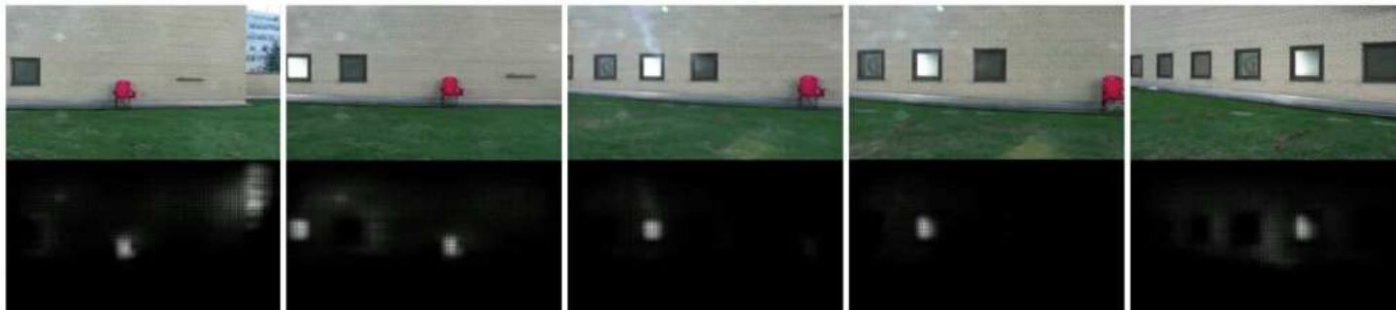
Liquid NCPs maintains its focus to complete the task



- A flight case study where an LSTM agent loses its target and gets confused by an adversary. Left to right shows the progress time.

The Evolution of LNNs: From Worms to ODEs

C LSTM gets confused by the window reflections

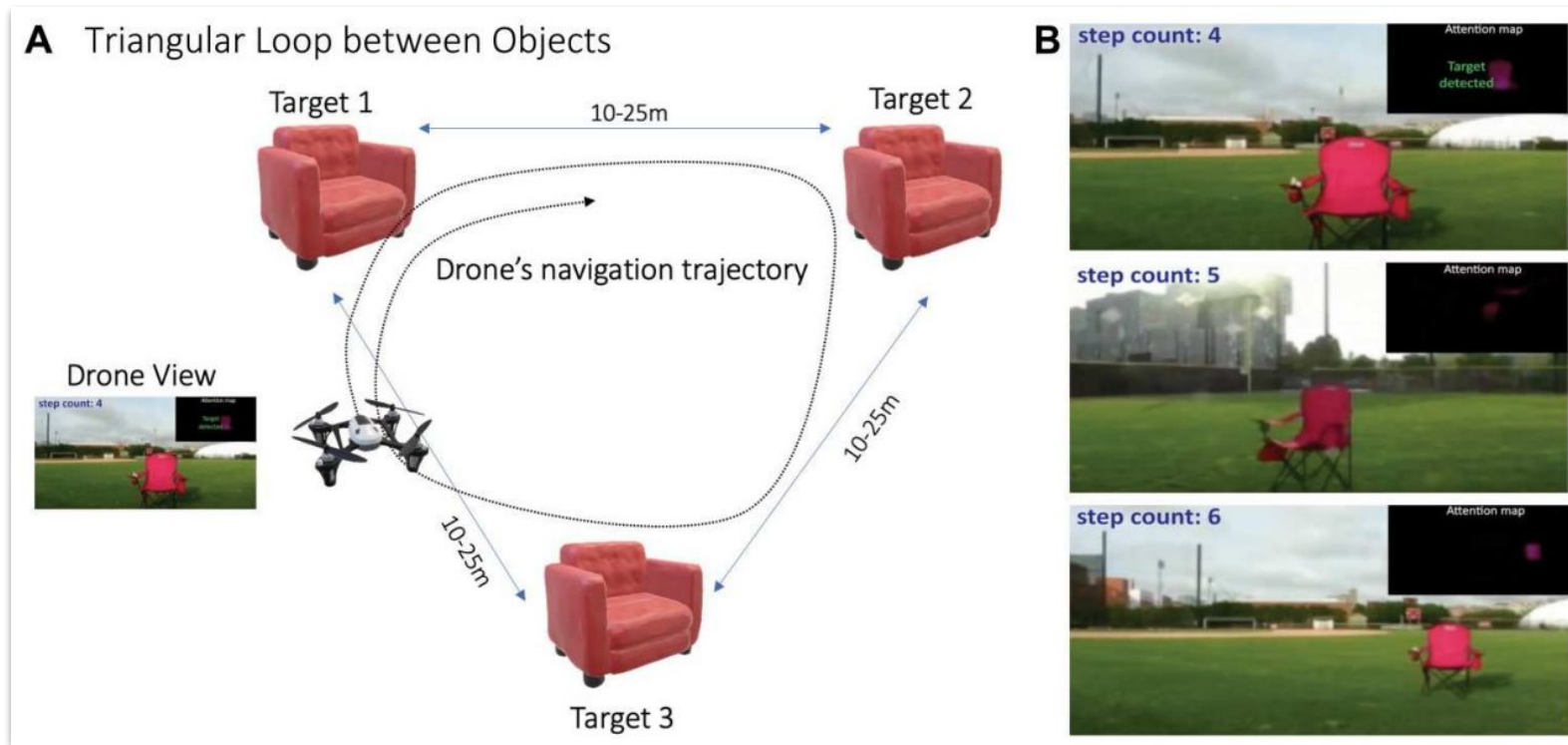


Liquid NCPs maintains its focus to complete the task



- A similar scenario as in (B), in a different OOD testing environment.

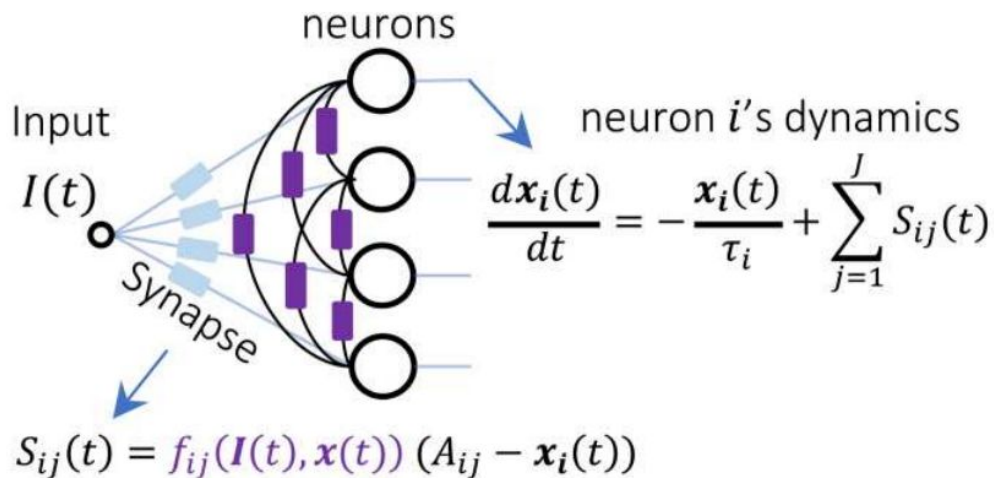
The Evolution of LNNs: From Worms to ODEs



- Triangular loop between objects. (A) The triangular infinite loop task setup. (B) Time instances of drone view together with their attention map at steps 4, 5, and 6 of the infinite loop

The Evolution of LNNs: From Worms to ODEs

A liquid time-constant network (LTC)

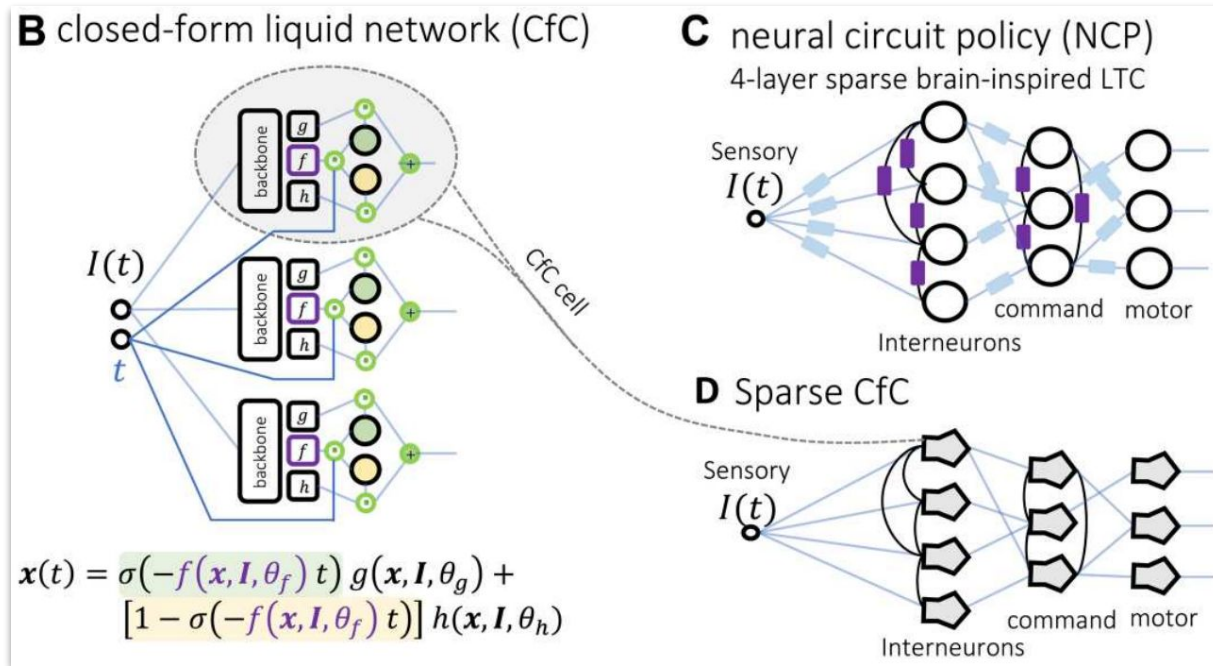


Legend

-  Liquid input synapse
-  Liquid synapse
-  Hadamard product
-  Sigmoidal gate (σ)
-  $1 - \sigma$
-  CfC Cell

- Liquid neural networks. (A) Schematic demonstration of a fully connected LTC layer
- The dynamics of a single neuron i is given, where $x_i(t)$ represents the state of the neuron i and τ represents the neuronal time constant. Synapses are given by a nonlinear function, where $f(\cdot)$ represents a sigmoidal nonlinearity and A_{ij} is a reversal potential parameter for each synapse from neurons j to i .

The Evolution of LNNs: From Worms to ODEs



- (B) Schematic representation of a closed-form liquid network (CfC) together with its state equation. (C) A schematic representation of an NCP. (D) CfC. A sparse neural circuit with the same architectural motifs as NCPs, this time with CfC neural dynamics.

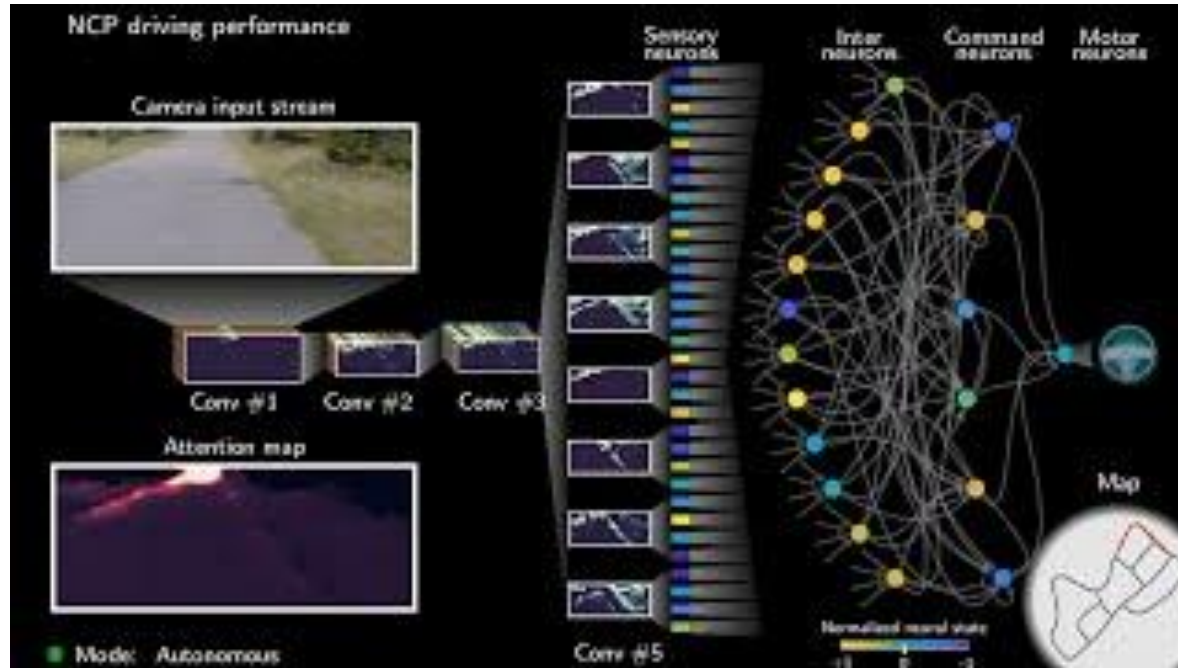
The Evolution of LNNs: From Worms to ODEs



- Drones navigate unseen environments with liquid neural networks

Sources: https://www.youtube.com/watch?v=RPFbk6D_dNw&t=48s

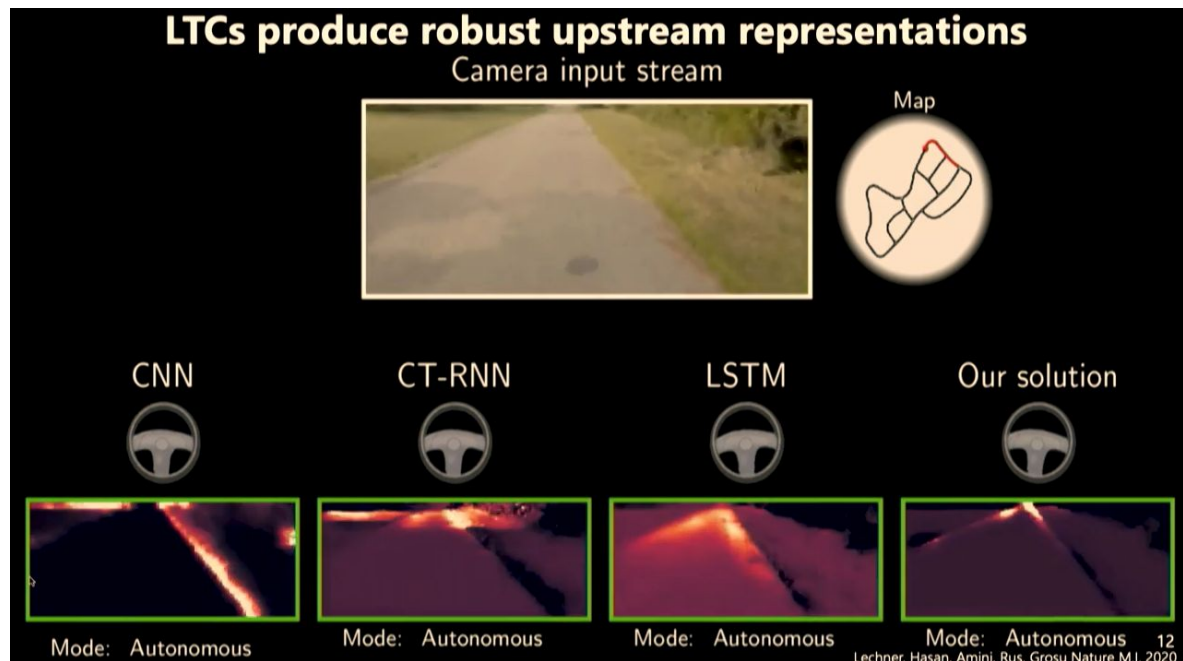
The Evolution of LNNs: From Worms to ODEs



- We have to consider many factors
- Car driving mechanics
- Road definition
- Artificial sensors
- Collision detection
- Traffic simulation
- Live stream variant
- Modeling
- Fine-tuning etc.

Sources: <https://www.youtube.com/watch?v=8KBO7NJh4Y>

The Evolution of LNNs: From Worms to ODEs



- Check attention maps
- For a system to work in a real world it must be causal

Sources: <https://www.youtube.com/watch?v=0FNkrjVlcuk>

The Evolution of LNNs: From Worms to ODEs



Sources: <https://www.youtube.com/watch?v=0FNkrjVlcuk>

- ဒီ talk မှာက CNN ရဲ့ attention နဲ့ LNNs ရဲ့ attention မှာ ဘယ်လို ကွာတယ်ဆိုတာကို မြှင်ရလိမ့်မယ်
- ပြီးတော့ autonomous flight simulation, Cheetah simulation တွေနဲ့ ပတ်သက်တာပါတယ်
- Proposal နဲ့ ပတ်သက်တာလည်း ရှင်ပြထားတယ်

The Evolution of LNNs: From Worms to ODEs

 > cs > arXiv:2504.02352  [Help](#)

Computer Science > Information Theory

[Submitted on 3 Apr 2025]

Liquid Neural Networks: Next-Generation AI for Telecom from First Principles

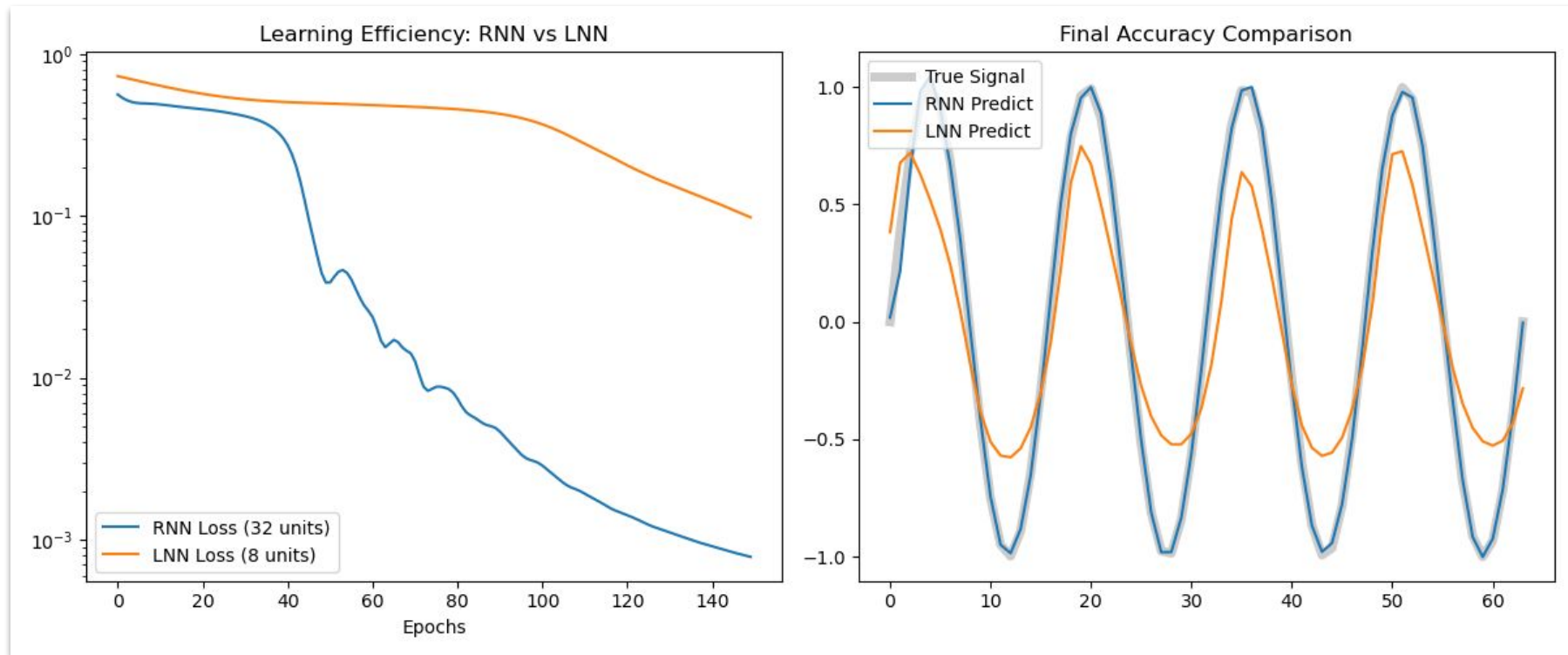
REQUEST CODE



Fenghao Zhu, Xinqian Wang, Chen Zhu, Chongwen Huang

Artificial intelligence (AI) has emerged as a transformative technology with immense potential to reshape the next-generation of wireless networks. By leveraging advanced algorithms and machine learning techniques, AI offers unprecedented capabilities in optimizing network performance, enhancing data processing efficiency, and enabling smarter decision-making processes. However, existing AI solutions face significant challenges in terms of robustness and interpretability. Specifically, current AI models exhibit substantial performance degradation in dynamic environments with varying data distributions, and the black-box nature of these algorithms raises concerns regarding safety, transparency, and fairness. This presents a major challenge in integrating AI into practical communication systems. Recently, a novel type of neural network, known as the liquid neural networks (LNNs), has been designed from first principles to address these issues. In this paper, we explore the potential of LNNs in telecommunications. First, we illustrate the mechanisms of LNNs and highlight their unique advantages over traditional networks. Then we unveil the opportunities that LNNs bring to future wireless networks. Furthermore, we discuss the challenges and design directions for the implementation of LNNs. Finally, we summarize the performance of LNNs in two case studies.

LSTM vs LNNs



Reference link: <https://www.kaggle.com/code/newtonbaba12345/liquid-neural-networks>

LSTM vs LNNs

- The LSTM uses three main gates:
 1. Forget Gate: Decides what old memory to throw away.
 2. Input Gate: Decides what new information to add.
 3. Output Gate: Decides what part of the memory to show the world right now.
- The Problem: These gates only open and close at fixed intervals (Step 1, Step 2, Step 3). They don't understand "time" passed between steps.

LSTM vs LNNs

- The CfC uses a Time-Gate. Instead of being triggered by a "step," this gate is a function of actual time (t).
- The Difference: While an LSTM gate says "Remember this at Step 5," a CfC gate says "The data is arriving 2.5 seconds late, so I will adjust the 'flow' of the liquid to account for that delay".

LSTM vs LNNs

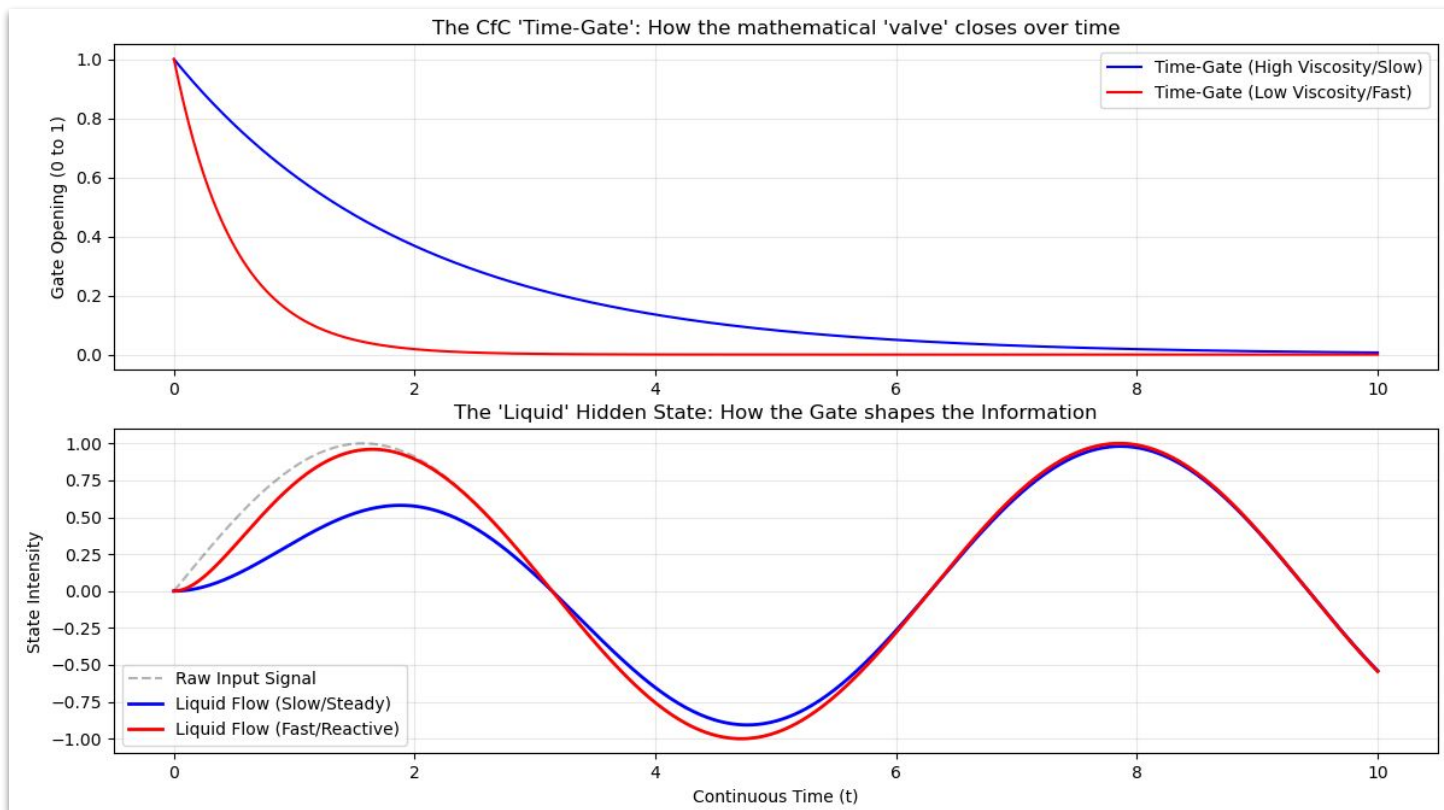
Feature	LSTM (Traditional)	LNN (Liquid)
Philosophy	Discrete: Steps in a ladder.	Continuous: Water in a pipe.
Logic	Gates (Switch on/off).	Differential Equations (Flow).
Flexibility	Breaks if time-steps change.	Adapts to any time-scale.
Complexity	Large (Many parameters).	Tiny (Highly efficient).

Closed-form Continuous-time (CfC)

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 def sigmoid(x):
5     return 1 / (1 + np.exp(-x))
6
7 def visualize_cfc_gates():
8     t = np.linspace(0, 10, 500)
9
10    # Simulate a signal (Information)
11    input_signal = np.sin(t)
12
13    # 1. The Time-Gate: exp(-t * tau)
14    # This represents the "Viscosity" of the liquid.
15    # Higher tau = faster decay/adaptation.
16    tau_slow = 0.5
17    tau_fast = 2.0
18
19    time_gate_slow = np.exp(-t * tau_slow)
20    time_gate_fast = np.exp(-t * tau_fast)
21
22    # 2. The Gating Mechanism: h(t) = G*gate + I*(1-gate)
23    # This is the "Closed-form" solution logic.
24    liquid_flow_slow = input_signal * (1 - time_gate_slow)
25    liquid_flow_fast = input_signal * (1 - time_gate_fast)
```

- Visualization of CfC gates

Closed-form Continuous-time (CfC)



Closed-form Continuous-time (CfC)

```
11 import tensorflow as tf
12 from ncps.tf import LTC, CfC
13 from ncps import wirings
14 import numpy as np
15 import time
16 import matplotlib.pyplot as plt
17
18 # 1. Setup Data
19 N = 100
20 t = np.linspace(0, 10, N)
21 x = np.sin(t).reshape(1, N, 1).astype(np.float32)
22 y = np.cos(t).reshape(1, N, 1).astype(np.float32)
23
24 # 2. Build LTC (The "Old" Liquid)
25 wiring = wirings.FullyConnected(16, 1)
26 ltc_model = tf.keras.Sequential([
27     tf.keras.layers.InputLayer(input_shape=(None, 1)),
28     LTC(wiring, return_sequences=True)
29 ])
30 ltc_model.compile(optimizer='adam', loss='mse')
31
```

- Python library
Neural Circuit
Policies (NCPs)
- Speed
comparison code
for Liquid
Time-Constants
(LTC) and CfC

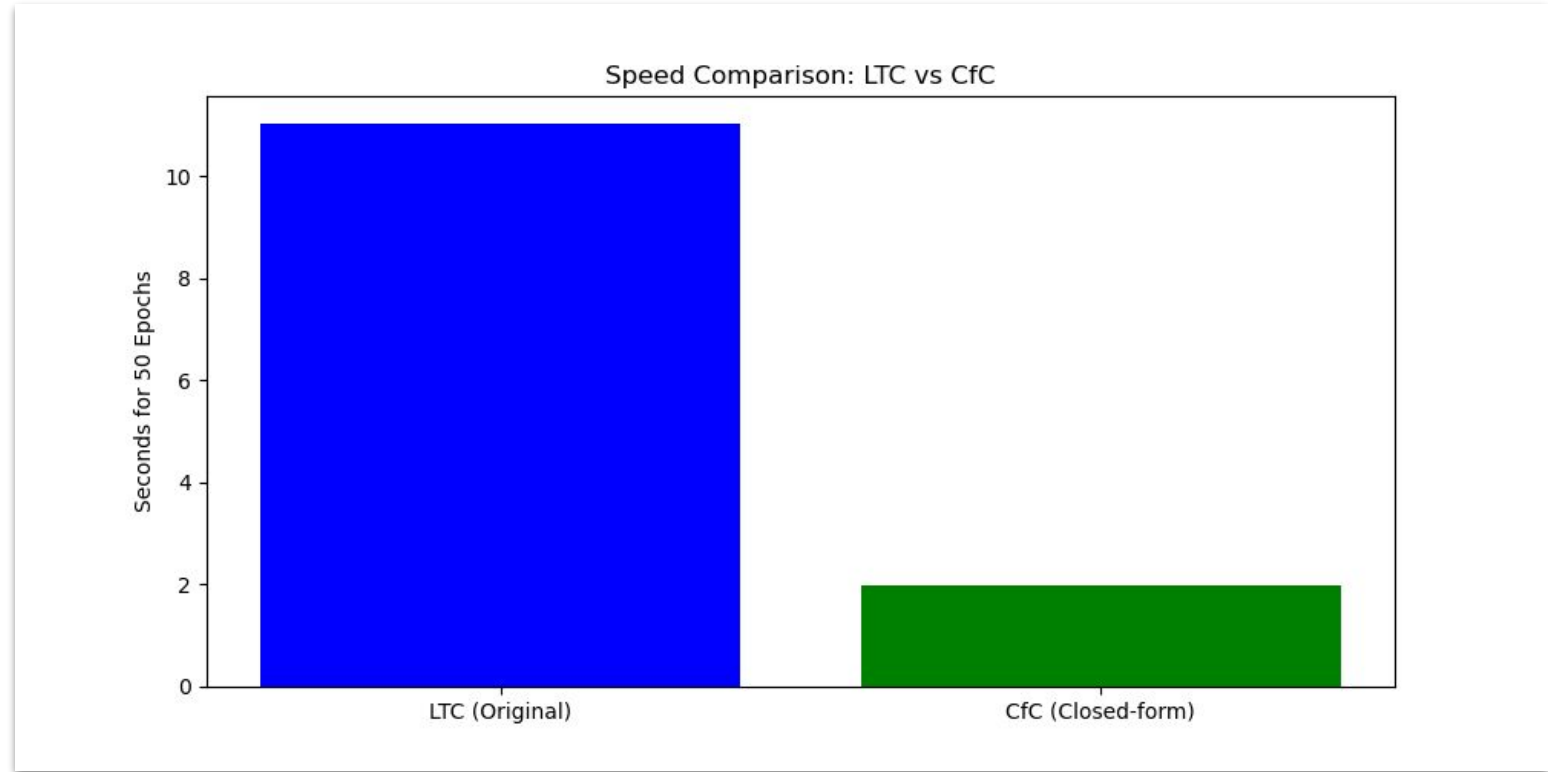
Closed-form Continuous-time (CfC)

```
32 # 3. Build CfC (The "New" Liquid)
33 cfc_model = tf.keras.Sequential([
34     ... tf.keras.layers.InputLayer(input_shape=(None, 1)),
35     ... # CfC is a drop-in layer. No complex wiring required by default.
36     ... Cfc(16, return_sequences=True)
37 ])
38 cfc_model.compile(optimizer='adam', loss='mse')
39
40 # 4. Benchmarking Speed
41 start = time.time()
42 ltc_model.fit(x, y, epochs=50, verbose=0)
43 ltc_time = time.time() - start
44
45 start = time.time()
46 cfc_model.fit(x, y, epochs=50, verbose=0)
47 cfc_time = time.time() - start
48
49 print(f"LTC (ODE-based) Training Time: {ltc_time:.2f}s")
50 print(f"CfC (Closed-form) Training Time: {cfc_time:.2f}s")
51
```

Closed-form Continuous-time (CfC)

```
52 # 5. Visualizing Architecture
53 # Note: CfC can be 'wired', but by default it acts as a high-speed continuous layer.
54 plt.figure(figsize=(10, 5))
55 plt.bar(['LTC (Original)', 'CfC (Closed-form)'], [ltc_time, cfc_time], color=['blue',
    'green'])
56 plt.ylabel("Seconds for 50 Epochs")
57 plt.title("Speed Comparison: LTC vs CfC")
58 plt.show()
```

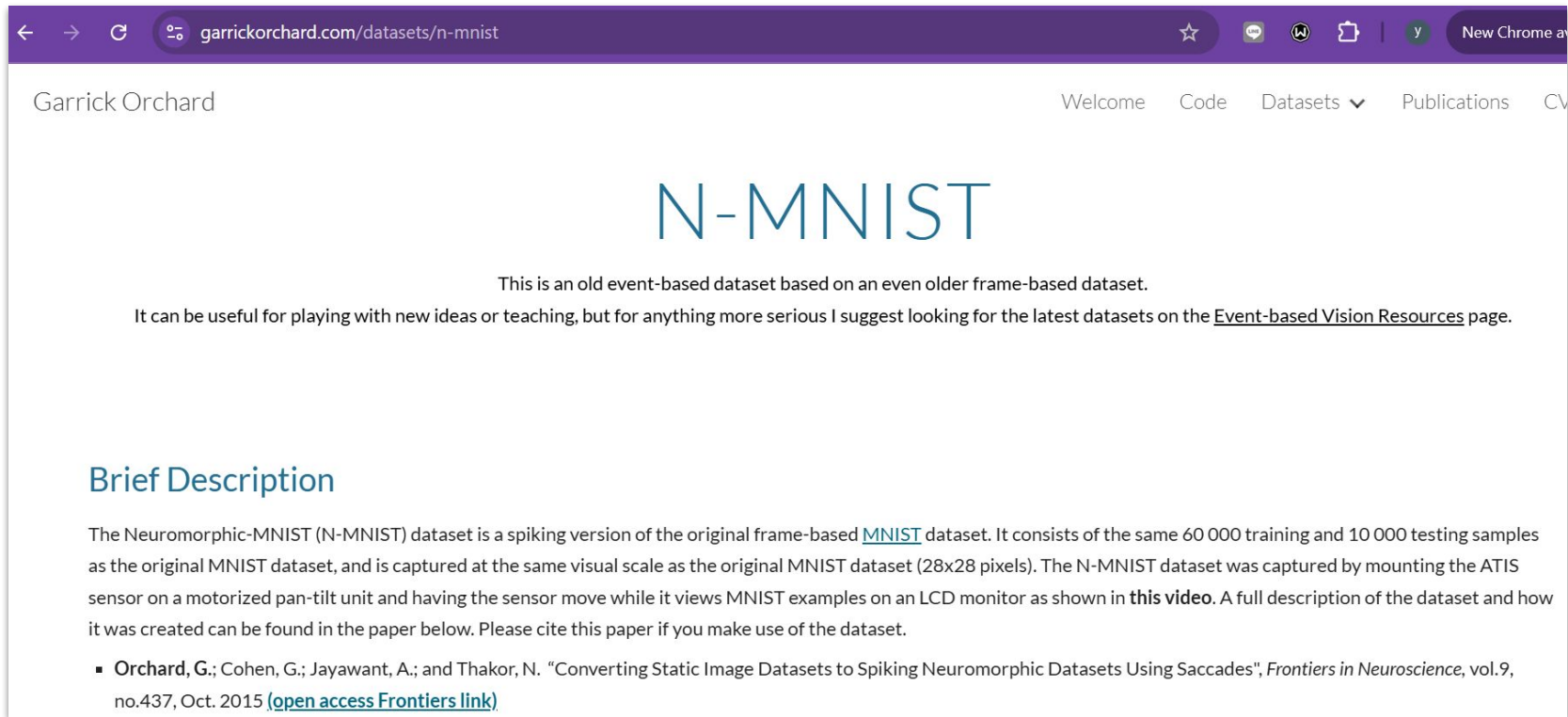
Closed-form Continuous-time (CfC)



Sequential Pattern Recognition with LNNs

- We conduct a large-scale benchmarking study across three distinct sequential modalities:
 1. Neuromorphic event-based data (N-MNIST)
 2. Stroke-based drawing coordinates (Google QuickDraw)
 3. Visual handwriting recognition (IAM Dataset)

Sequential Pattern Recognition with LNNs

A screenshot of a web browser showing the 'N-MNIST' dataset page by Garrick Orchard. The browser's address bar shows 'garrickorchard.com/datasets/n-mnist'. The page has a purple header with navigation links: 'Welcome', 'Code', 'Datasets' (with a dropdown arrow), 'Publications', and 'CV'. The main content area has the title 'N-MNIST' in large blue letters. Below the title, a paragraph states: 'This is an old event-based dataset based on an even older frame-based dataset. It can be useful for playing with new ideas or teaching, but for anything more serious I suggest looking for the latest datasets on the [Event-based Vision Resources](#) page.' A section titled 'Brief Description' follows, explaining that the Neuromorphic-MNIST (N-MNIST) dataset is a spiking version of the original frame-based MNIST dataset, consisting of 60,000 training and 10,000 testing samples. It was captured by mounting the ATIS sensor on a motorized pan-tilt unit. A full description and citation are provided at the bottom.

garrickorchard.com/datasets/n-mnist

Garrick Orchard

Welcome Code Datasets ▾ Publications CV

N-MNIST

This is an old event-based dataset based on an even older frame-based dataset.

It can be useful for playing with new ideas or teaching, but for anything more serious I suggest looking for the latest datasets on the [Event-based Vision Resources](#) page.

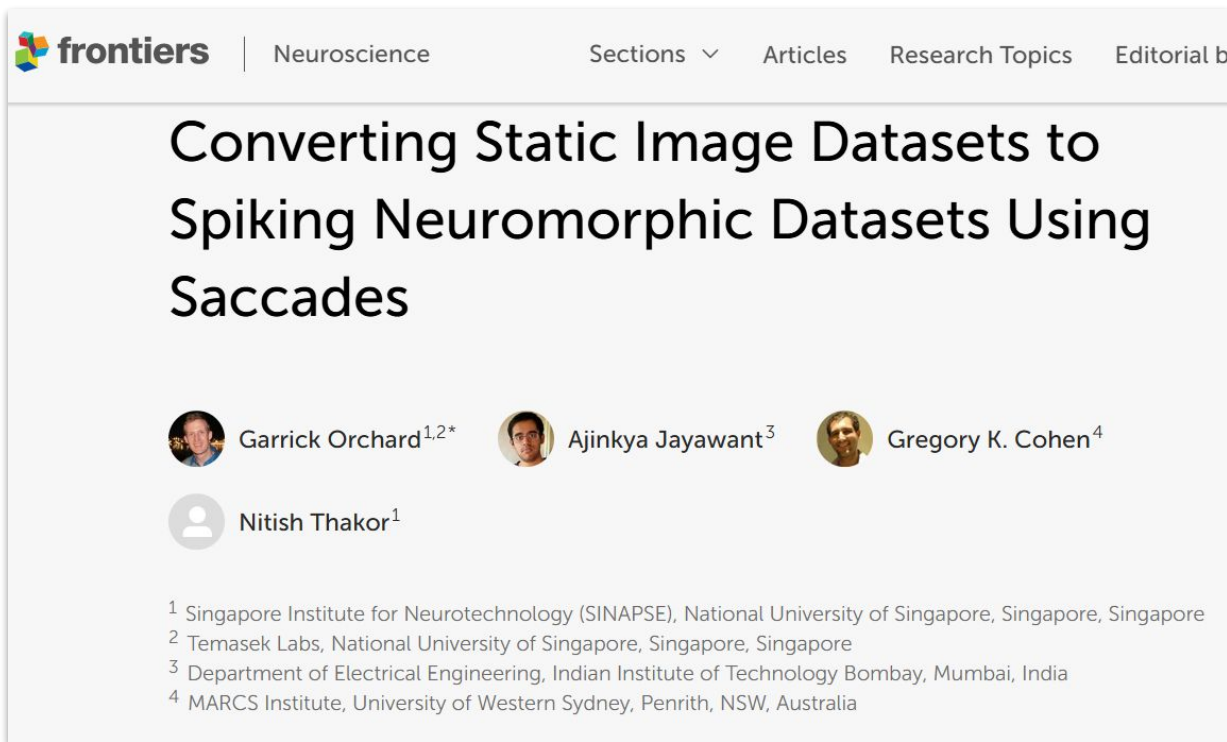
Brief Description

The Neuromorphic-MNIST (N-MNIST) dataset is a spiking version of the original frame-based [MNIST](#) dataset. It consists of the same 60 000 training and 10 000 testing samples as the original MNIST dataset, and is captured at the same visual scale as the original MNIST dataset (28x28 pixels). The N-MNIST dataset was captured by mounting the ATIS sensor on a motorized pan-tilt unit and having the sensor move while it views MNIST examples on an LCD monitor as shown in [this video](#). A full description of the dataset and how it was created can be found in the paper below. Please cite this paper if you make use of the dataset.

- Orchard, G.; Cohen, G.; Jayawant, A.; and Thakor, N. "Converting Static Image Datasets to Spiking Neuromorphic Datasets Using Saccades", *Frontiers in Neuroscience*, vol.9, no.437, Oct. 2015 ([open access Frontiers link](#))

Source: <https://www.garrickorchard.com/datasets/n-mnist>

Sequential Pattern Recognition with LNNs



- 2015 လောက်က စာတမ်းပါ
- နာမည်ကြီး Static ပုံဒေတာတွေ ဖြစ်တဲ့ MNIST နဲ့ Caltech101 ကို neuromorphic vision dataset အဖြစ် ပြောင်းတဲ့ စာတမ်း

Source: <https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2015.00437/full>

Sequential Pattern Recognition with LNNs



FIGURE 2 | (A) A picture of the ATIS mounted on the pan tilt unit used in the conversion system. **(B)** The ATIS placed viewing the LCD monitor.

Source: <https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2015.00437/full>

Sequential Pattern Recognition with LNNs

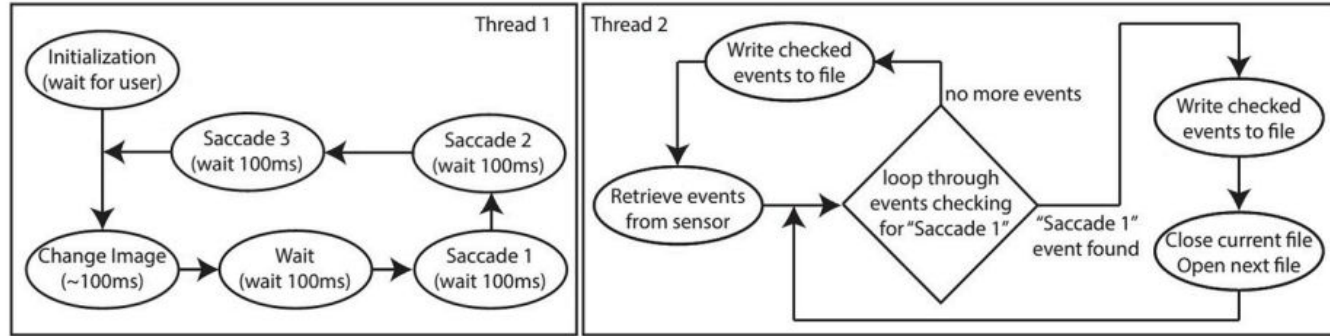
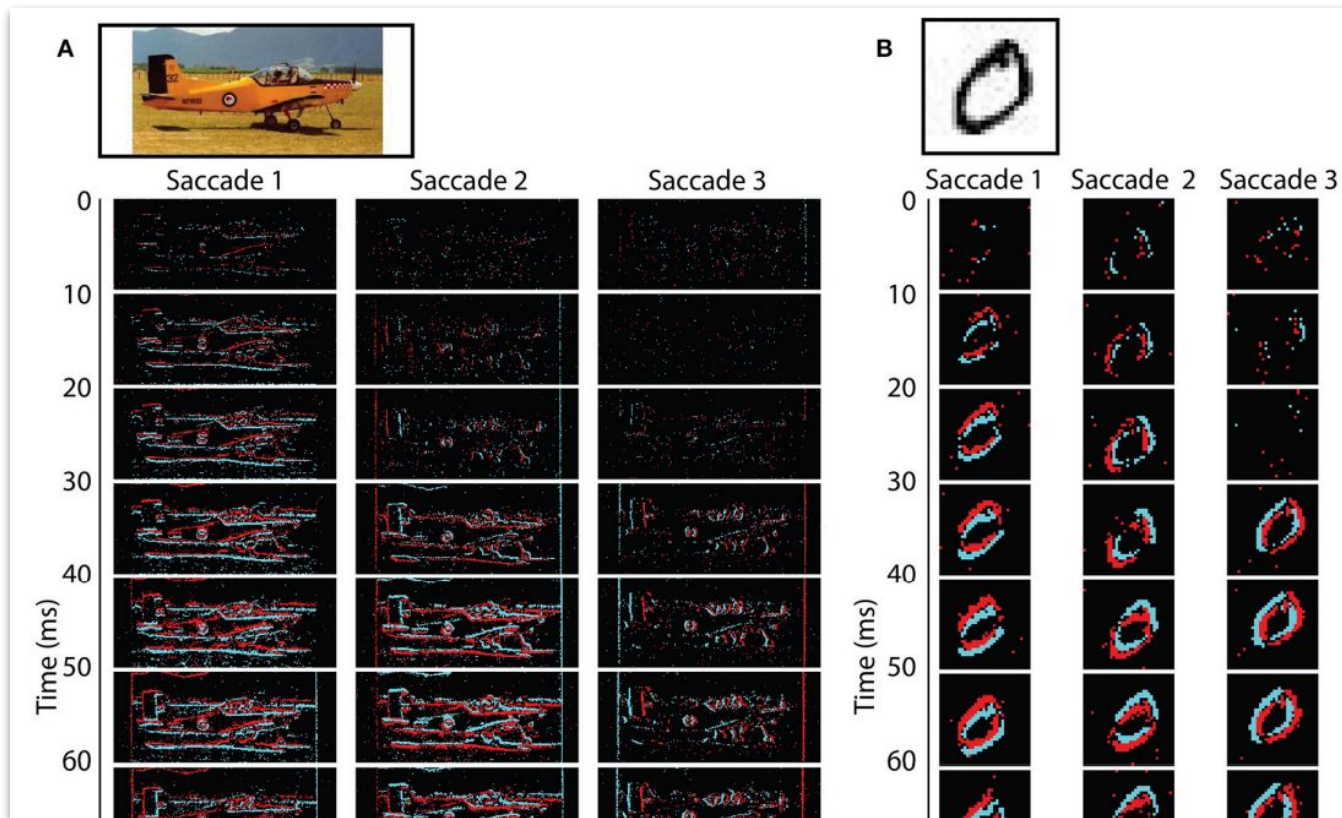


FIGURE 3 | Software flow diagram of the PC code showing two threads. Thread 1 (left) controls the timing of microsaccades and changing of images on the screen, while thread 2 (right) controls acquisition of the event data and writing it to disk.

Source: <https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2015.00437/full>

Sequential Pattern Recognition with LNNs



Sequential Pattern Recognition with LNNs

Quick, Draw! The Data

 Get the data

 Play

Select a drawing



Sequential Pattern Recognition with LNNs

You are looking at 103,031 cat drawings made by real people... on the internet.

If you see something that shouldn't be here, simply select the drawing and click the flag icon.

It will help us make the collection better for everyone.



Source: <https://quickdraw.withgoogle.com/data>

Sequential Pattern Recognition with LNNs

(i) Start from the first column, ask the following question:

the Prime Minister to Paris was derppped

The wish was particularly apparent in comments

The team is composed of experienced

Source: <https://github.com/tuandoan998/Handwritten-Text-Recognition>

Sequential Pattern Recognition with LNNs

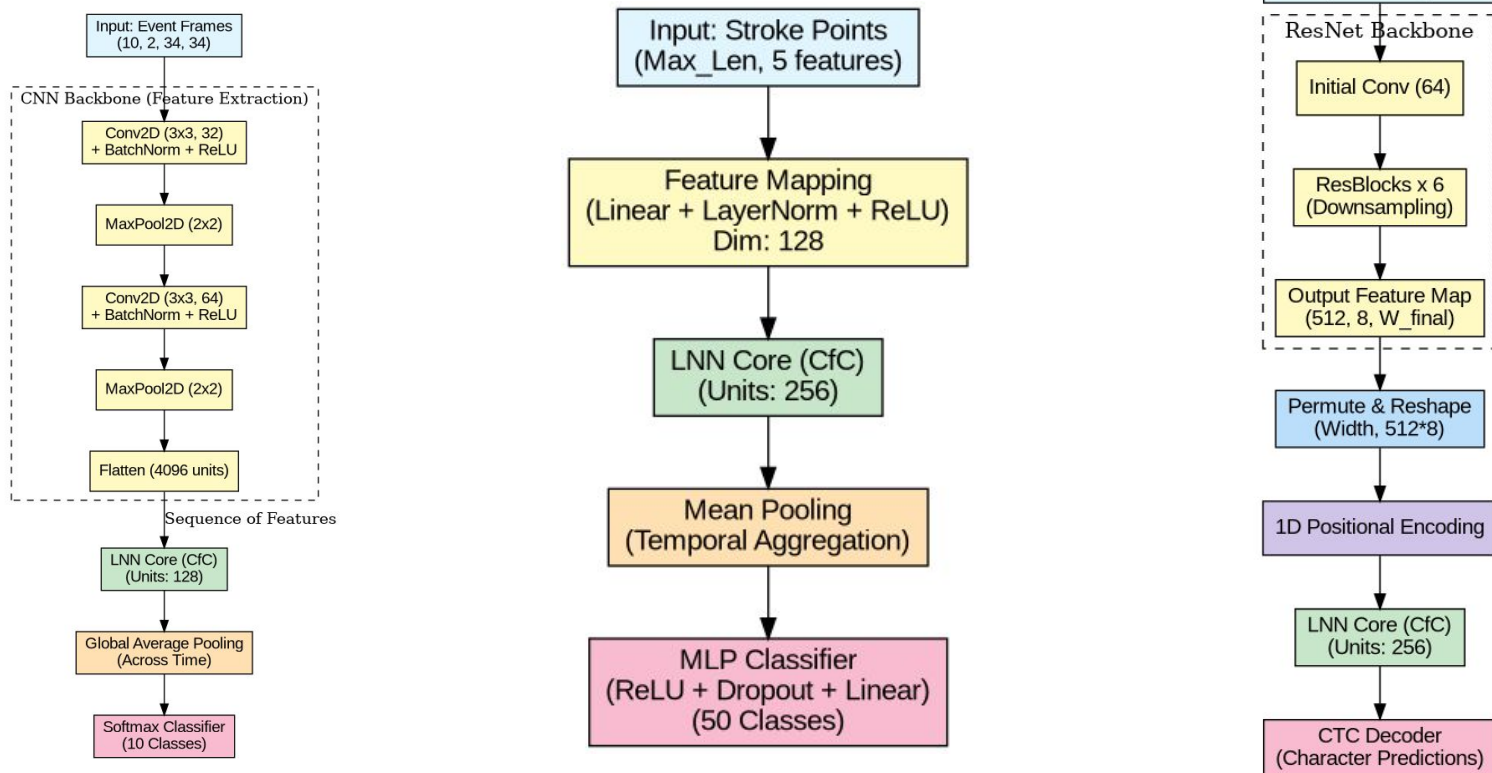


Fig. Architecture for N-MNIST, QuickDraw and IAM

Sequential Pattern Recognition with LNNs

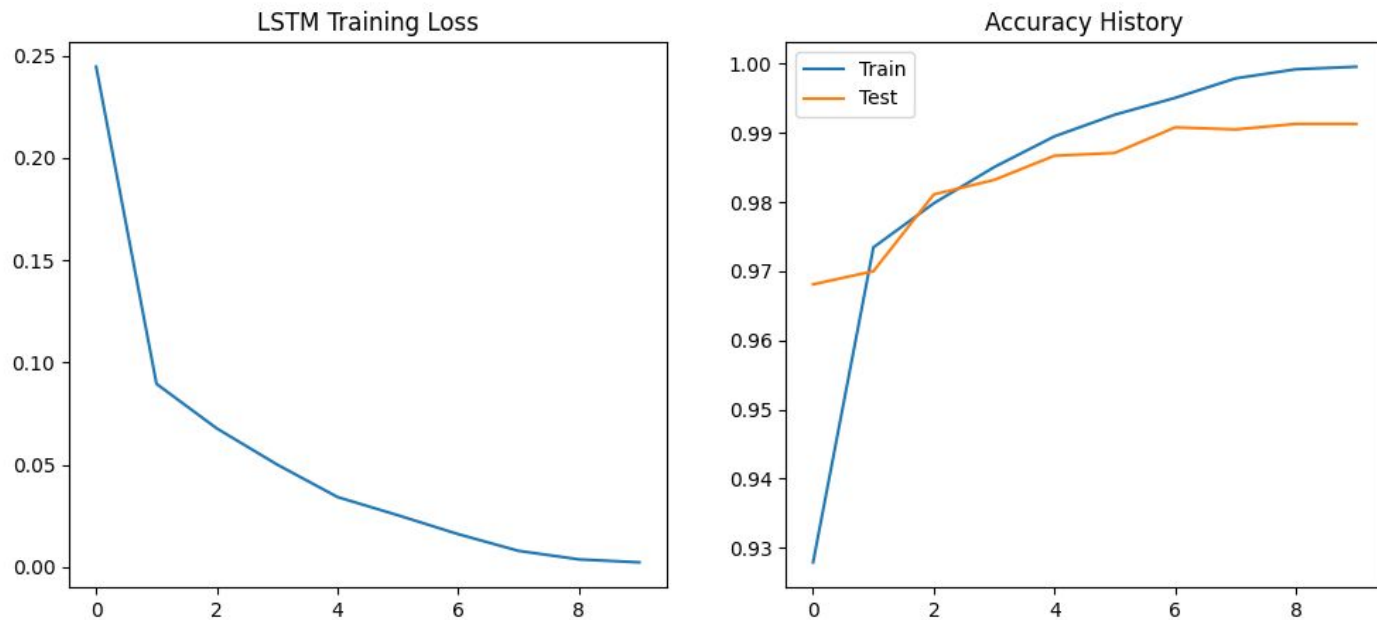


Fig. N-MNIST LSTM training loss and accuracy graph

Sequential Pattern Recognition with LNNs

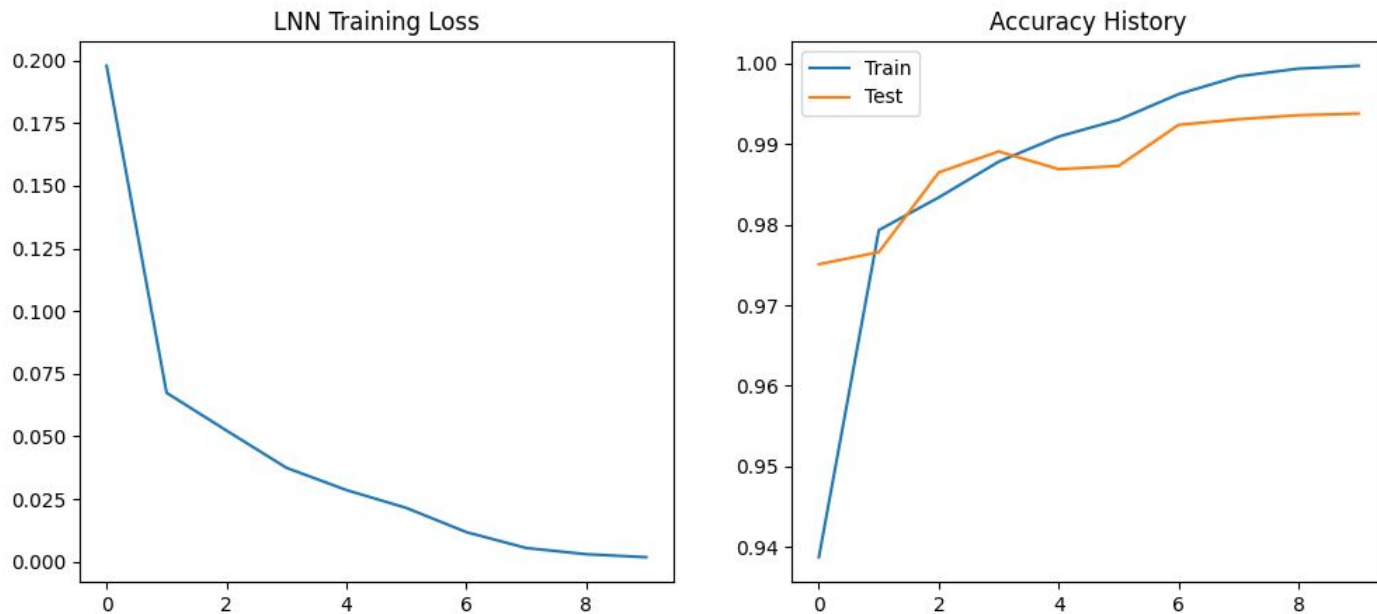


Fig. N-MNIST LNNs training loss and accuracy graph

Sequential Pattern Recognition with LNNs

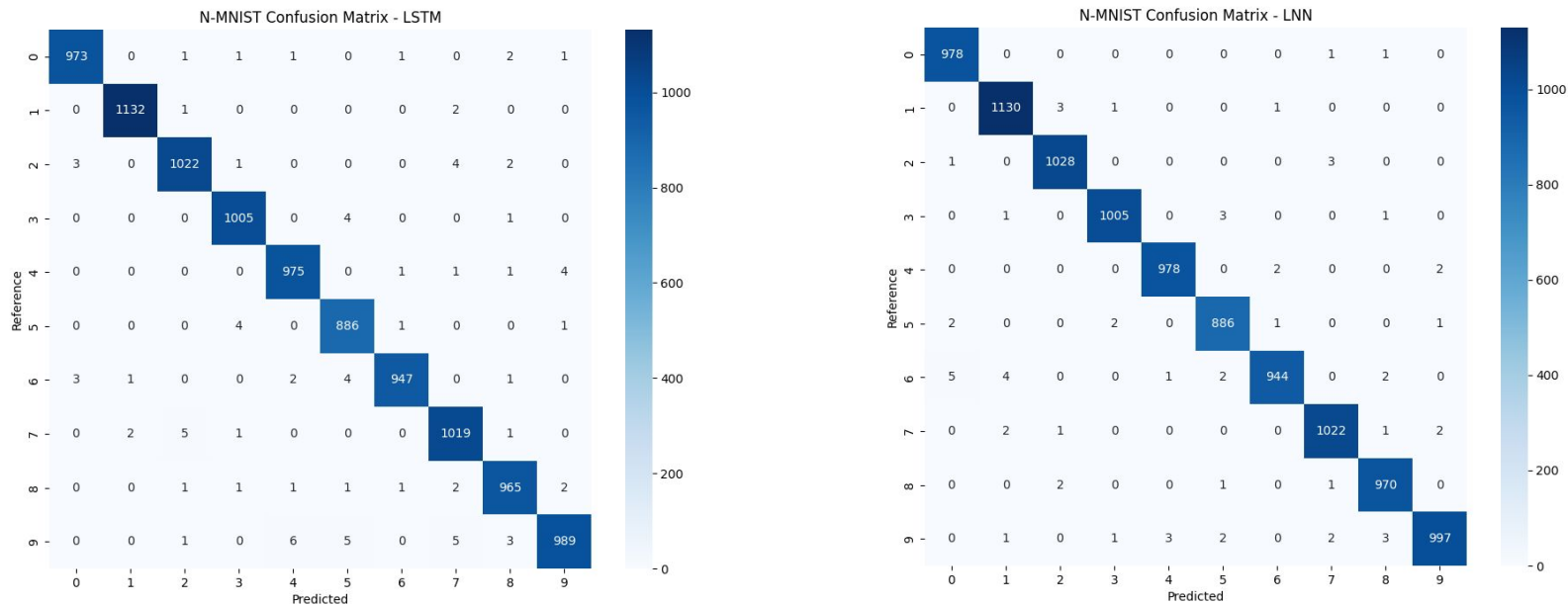


Fig. Confusion matrix for N-MNIST dataset, Left: LSTM, Right: LNNs

Sequential Pattern Recognition with LNNs

Dataset	Model	Metric	Train	Test
N-MNIST	LNN	Acc	0.9997	0.9938
	LSTM	Acc	0.9996	0.9913
QuickDraw	LNN	Acc	0.9978	0.9577
	LSTM	Acc	1.0000	0.9701
IAM	LNN	CER	0.1717	0.1237
	LSTM	CER	0.0784	0.1090

Table. Quantitative Benchmark: LNNs vs. LSTM

Fine-tuning with Liquid Foundation Model

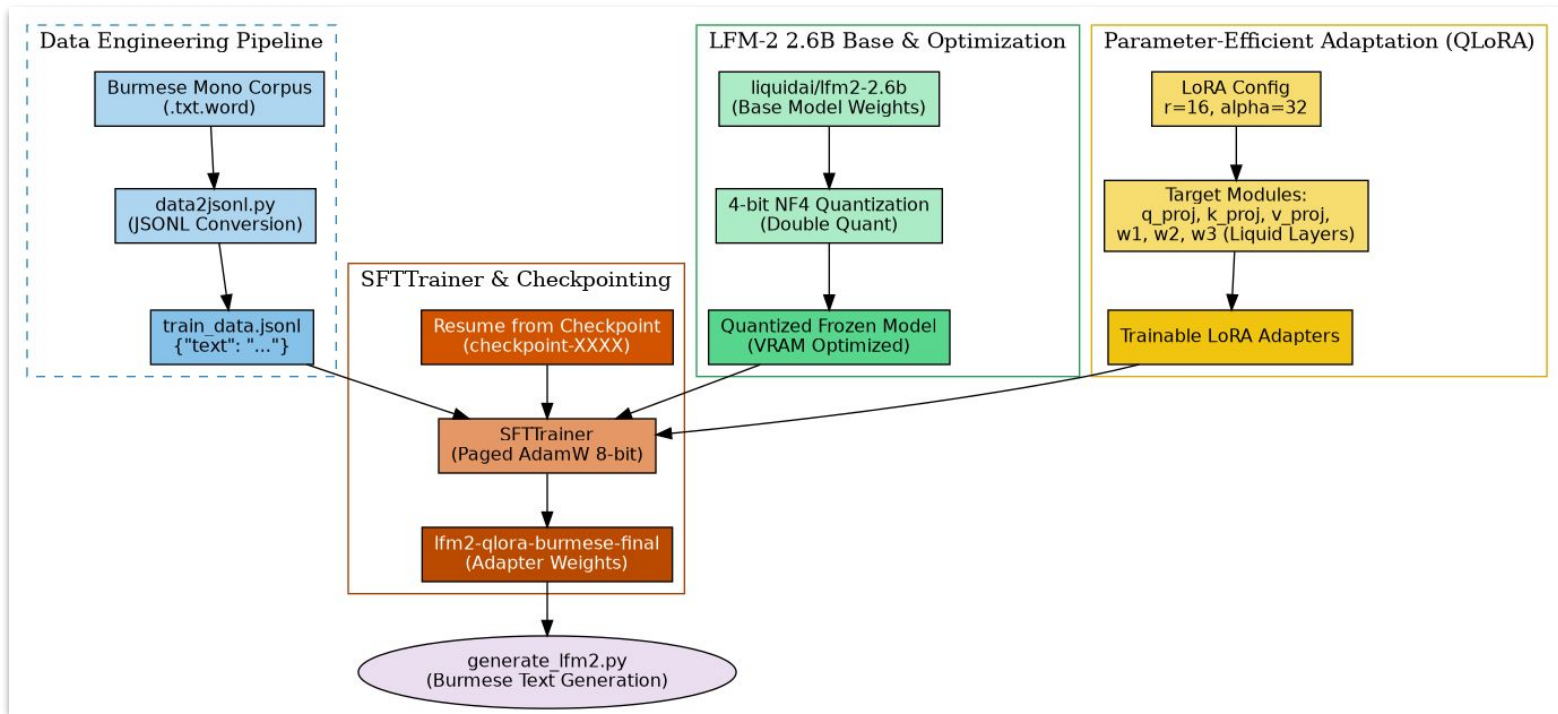


Fig. Overview of Supervised Fine-Tuning with LFM-2-2 6B Base model

Fine-tuning with Liquid Foundation Model

```
ye@lst-hpc3090: ~/exp/myMono_LFM
{'loss': 0.5701, 'grad_norm': 2.3793764114379883, 'learning_rate': 1.3634194559956373e-08, 'entropy': 0.6031446309760213, 'num_tokens': 83132603.0, 'mean_token_accuracy': 0.8193988040089607, 'epoch': 1.0}
{'loss': 0.5564, 'grad_norm': 2.2205424308776855, 'learning_rate': 1.2498011679960007e-08, 'entropy': 0.5761293834075332, 'num_tokens': 83149601.0, 'mean_token_accuracy': 0.8268802650272846, 'epoch': 1.0}
{'loss': 0.5312, 'grad_norm': 2.6596286296844482, 'learning_rate': 1.1361828799963643e-08, 'entropy': 0.5508037943392992, 'num_tokens': 83166386.0, 'mean_token_accuracy': 0.8247018188238144, 'epoch': 1.0}
{'loss': 0.5234, 'grad_norm': 2.1411092281341553, 'learning_rate': 1.022564591996728e-08, 'entropy': 0.5520416988059879, 'num_tokens': 83183952.0, 'mean_token_accuracy': 0.8335484981536865, 'epoch': 1.0}
{'loss': 0.5497, 'grad_norm': 2.423283576965332, 'learning_rate': 9.089463039970914e-09, 'entropy': 0.5823555167764425, 'num_tokens': 83200428.0, 'mean_token_accuracy': 0.8191213145852089, 'epoch': 1.0}
{'loss': 0.5564, 'grad_norm': 2.0099804401397705, 'learning_rate': 7.95328015997455e-09, 'entropy': 0.5699185688048601, 'num_tokens': 83218990.0, 'mean_token_accuracy': 0.8298507906496525, 'epoch': 1.0}
{'loss': 0.5235, 'grad_norm': 1.9973183870315552, 'learning_rate': 6.817097279978186e-09, 'entropy': 0.551297714188695, 'num_tokens': 83236330.0, 'mean_token_accuracy': 0.8331429354846478, 'epoch': 1.0}
{'loss': 0.5467, 'grad_norm': 2.3594107627868652, 'learning_rate': 5.680914399981822e-09, 'entropy': 0.5545737722888588, 'num_tokens': 83254701.0, 'mean_token_accuracy': 0.8280885197222233, 'epoch': 1.0}
{'loss': 0.572, 'grad_norm': 2.245898962020874, 'learning_rate': 4.544731519985457e-09, 'entropy': 0.5783307179808617, 'num_tokens': 83272339.0, 'mean_token_accuracy': 0.8227200582623482, 'epoch': 1.0}
{'loss': 0.578, 'grad_norm': 2.3958323001861572, 'learning_rate': 3.408548639989093e-09, 'entropy': 0.5950157677754759, 'num_tokens': 83287288.0, 'mean_token_accuracy': 0.8191494919359684, 'epoch': 1.0}
{'loss': 0.5632, 'grad_norm': 2.3703255653381348, 'learning_rate': 2.2723657599927285e-09, 'entropy': 0.5862594863399864, 'num_tokens': 83304363.0, 'mean_token_accuracy': 0.8243992872536182, 'epoch': 1.0}
{'loss': 0.5549, 'grad_norm': 2.2291901111602783, 'learning_rate': 1.1361828799963642e-09, 'entropy': 0.5732654126361012, 'num_tokens': 83320591.0, 'mean_token_accuracy': 0.8250299856066704, 'epoch': 1.0}
{'train_runtime': 67356.924, 'train_samples_per_second': 20.907, 'train_steps_per_second': 1.307, 'train_loss': 0.1620903710070505, 'entropy': 0.5838482262658291, 'num_tokens': 83333483.0, 'mean_token_accuracy': 0.8219176960773156, 'epoch': 1.0}
100%
88014/88014 [18:42:36<00:00, 1.31it/s]

real    1123m9.762s
user    1044m57.858s
sys      74m44.848s
(lfm_env) ye@lst-hpc3090:~/exp/myMono_LFM$
In total 740m + 1123m.
```

Fig. Fine-tuning Log (about 31 hours)

Fine-tuning with Liquid Foundation Model

1 ပြဇာတ် ရဲ့ ဆွဲဆောင် မှု ဟာ စကားလုံး တွေ နဲ့ သရုပ်ဆောင် ပေါ် မှာ မှတည် တယ်။ ဒီဗျူ အပိုင် ဂီတ ၃၀ မိ မှု (Kidney
2 သူ တို့ က ဘတ်စ်ကား စီး ကြ တယ်။ ရှေ့ အလွမ်းဝံ့ ဟန္တ ပါဦး မည် ဖ
3 ရာသီဥတု ခန့်မှန်းချက် က ပြော တယ်။ မကြာခင် လေ မုန်တိုင်း တိုက် တော့ မယ်။ ဟူ၏။ အဆွေ ဖါ (အညံ့ထိုင် ၃)
4 တတ် နိုင် ရင် မြန် နိုင် သမျှ အမြန်ဆုံး လာ ပေး ပါ။ (ဘ) ၃၂၇၉ - ဝန် ဒီ ယူ (၄ . ၇ . ထွက်) ဖ
5 ကျွန်တော် တို့ ရေတံခွန် ကို လမ်းလျှောက် သွား လို့ ရ လား။ ဘင်္ဂ ဝ၏ ထပ် ဖြစ် ရ မည် ဟူ၍ ဒါဟာ
6 ဆွေမျိုး တွေ ဆီ လာလည် တာ ပါ။ ရှင်သား အနေး (ဖ) ၂ , ၃ စ ခ ကြည့်ဝ
7 နှင် ဈေးဝယ် ငါ ဟင်းချက် မယ်။ ပြာ တို့ ဘီ သူနဲ့ အလူးလဲ ရှိ စံ
8 ကိုယ်ခန္ဓာ အရမ်း ပင်ပန်း နေ တဲ့ အချိန် မှာ ဆူညံ တဲ့ သီချင်းသံ တောင် မှ သားချောတေး လို ကြား ရ တယ်။ ဟာ စောဖတ်
9 ခင်ဗျား အကြွေးဝယ်ကဒ် လက်ခံ လား။ မုန်ရှန်ပါဏ္ဍ တိုက်သည် ဆူးစီ ထ
10 နှင် ဈေးဝယ် ငါ ဟင်းချက် မယ်။ လူဒီရုံ သားပြီ အဖွဲ့ တို့၏ ဆစ္စ
11 ၁၉၄၈ ခုနှစ် ၊ ဇွန်ဝါရီ လ ၄ ရက်နေ့ တွင် ဗြိတိသျှ အင်ပါယာ ထံ မှ နိုင်ငံ ၏ လွတ်လပ် ရေး ကို ပြန်လည် ရရှိ ခဲ့ သည် နှင့်
12 ဖြစ် လာ ခဲ့ ပါ သည်။ (၂) ၃ - ၆ : ၁ - ၀ . (ရှိ , ၆ , ၄ , ၅ - ၈ - ၆ -
13 အိတ် က မီးရထား ပေါ် မှာ ကျန် နေ ခဲ့ တာ ပါ။ ဒု ဖြစ် လုံ ယွင်း၏ သဘား ဝန် ဗုဒ္ဓ
14 ကျွန်တော် အဲဒါ ကို တွေ့ ခဲ့ ရင် အဲဒီ မှာ ရပ် ပြီး တော့ ဂျူဒီ ရဲ့ အိမ် သွား တဲ့ လမ်း ကို မေး ခဲ့ ရ မှာ။ ထိမ်စံ ဘယ်ညို သ်
15 သူ ၏ အချိန်ကာလ အတွင်း တွင် ၁၉၉၇ အာရှ စီးပွား ရေး ပြဿနာ ၏ ရှိက်ခတ် မှု များ ၊ ၂၀၀၃ ခုနှစ် ဆားရောဂါ ပျံ့နှံ့ မှု ၊
16 ဂျမားအစ္စလာမိယား အဖွဲ့ ၏ အကြမ်းဖက် တိုက်ခိုက် မည် ခြိမ်းခြောက် မှု များ ကို ဖြေရှင်း ပေး ခဲ့ သည်။
17 အေး ဟုတ် ပါ တယ်။ ရာသီ နဲ့ ၆.၅ - မြင်း ဘူး () လိုင် ထံ
18 ပြင်ဦးလွင် မြို့နယ် သည် မန္တလေး တိုင်း ၊ ပြင်ဦးလွင် ခရိုင် အတွင်း တွင် တည်ရှိ သည်။ ဘာဝါ၌ ဂျူး ဆွဲ ၇ ၉ - စံ နီ ထက်
19 ခင်ဗျား ဆက် တာ မှား နေ ပြီ။ ဒါ ဝို့ သွေး ရယ် အလျူးဘီ တဲ့ ထည်
20 မျက်လုံး မိုတ် ပြီး အရင် အတိတ် အကြောင်း ကို တွေး နေ ခဲ့ တယ်။ သူ စဉ်၏ ၂ ၃ - ဦး () ဘဝ ဖြည့်ဆွဲ ကာစွ
21 ဒီ မှာ စောင့် ပါ။ အလိုးဘဝရင်တူ နဲ့ အဆွေ သည် ဖြတ်
22 သီးခြား အခန်း ၊ ဒါဟု ဝမ်၏ ပညာတိယောင် ဆွဲ လွဲ ထူး

- အစ စာကြောင်း အိုကေ
- ပုဒ်မ ရဲ့ နောက်မှာ ပြဿနာရှိ

Fig. Generated text with fine-tuned model

Fine-tuning with Liquid Foundation Model

Evaluation

```
(lfm_env) ye@lst-hpc3090:~/exp/myMono_LFM$ ./run_inference.sh
```

1. Running Perplexity Evaluation

Loading tokenizer and model...

```
torch_dtype is deprecated! Use `dtype` instead!
```

Evaluating on otest.1k.word...

100%

1000/1000 [00:27<00:00, 36.76it/s]

Evaluation Results

Average Loss: 0.9616

Perplexity: 2.6159

```
real    0m34.626s
```

```
user      0m37.286s
```

```
sys      0m0.640s
```

Fig. Perplexity for fine-tuned model

Fine-tuning with Liquid Foundation Model

1 ဘာဖြစ်လို့လဲ။ ။ ၊ ရင်ထပ်ဆေးသန့်တွေးအောင်း (T y p e)
 2 ခင်ဗျားကြီးစားရမယ်။ ။ ၈ - ၉ (သိ) = ၃ + ၆ . . ပါလဲ (အဖေ့တံ) ဆို
 3 သူငယ်ချင်းထံကကျောက်စိမ်းလည်ဆွဲကိုလက်ဆောင်အဖြစ်ရခဲ့တယ်။ ။ နှင်း ၊ ၃ - ၃ . ၈ ၉ ဒီ
 4 ၆ နာရီလောက်ကြာတယ်။ ။ " မိုးသား " ဟဲဘူး (အဖွဲ့ပါ) - . . . ရှိ
 5 ကုသမသည်ကုသိမ်ဖြစ်လျှင်လောကဓာတ်အရကနှင့်သတို့ရန်ဂျိဟ်ကျသဖြင့်မိတ်ဖက်ဖြစ်စေ
 6 သည်။ ။ ၃ ၆ % ထဲမှစီဒီယားဘဏ္ဍာဝတ် (R C) လ
 7 မည်သည့်အတိုက်အခံပါတီကမျှအရပ်သုံးဖြစ်သည့်ဗမာဆိုသည့်ဝေါဟာရကိုမြန်မာနိုင်ငံ
 8 ။ " ဒယ်လဲ (၃) လွှဲဘူးဂနိုင်တို့
 9 ကျွန်တော်ရထားပြောင်းဖို့လိုအပ်လား။ ။ မှာသူဟာဂံစီဒါဝယ် ၃ - ၄ % (ည) . .
 10 အခြားကုန်မြူနစ်များ၊ အထူးသဖြင့်သခင်စိုးနှင့်သခင်သိန်းဖေတို့သည်မြို့တံသျှတို့ဆုတ်ခွာ
 11 ခင်ကပင်ဂျပန်တော်လှန်ရေးတရားဟောခဲ့၏။ ။ ဒီဘယ်ဝါသဖြင့်အနွှေရနွှေကို
 12 ဆိုလိုသည်မှာမြေကြီးထဲမှ m i n i n g သမားတူးထုတ်ပေးပြီးဆိုလျှင်ကုန်ကြမ်းအဆင့်ထိ
 13 သည်။ ။ ဒါဘယ်ဟဲဦးဇဝန် (. ၊) . (2 - 3 ° C -) ဖွဲ့
 14 ခင်ဗျားတို့နိုင်ငံတအားတိုးတက်လာတာပဲ။ ။ ၊ - . . ၆ , မြေထွက်ဖူးရှယ်စီဒါသည်
 15 ဘယ်လောက်ပြောပြောနားမထောင်ဘူး။ ။ သံရှုတ်ဝိစီဂျင်အဖွဲ့နို့ဆီ
 16 ဘာအရောင်ရှာပါသလဲ။ ။ နိုးတွေမြင်းစီးဆိုဖူးများ (
 17 ကျွန်တော်လည်ချောင်းနာနေတယ်။ ။ ၈ မှ - ပြီသူ့အိုးစံဖိုးမြဟု
 18 ကောင်းပါပြီ။ ။ ဒီဟာကိုယူပါမယ်။ ။ ရှုံ့ဖွဲ့သတ္တဗျူးလည်း (K a s d y c e) အန်
 19 ဆေးနည်းနည်းစားလိုက်၊ သုံးလေးရက်လောက်အနားယူလိုက်ရင်ပျောက်သွားမှာပါ။ ။ ၊ . ၃ % ဟုတ်
 20 ဆိုလိုသည်မှာမြေကြီးထဲမှ m i n i n g သမားတူးထုတ်ပေးပြီးဆိုလျှင်ကုန်ကြမ်းအဆင့်ထိ
 သည်။ ။ ၈ - ၃ . ဝယ်ဗွန်ဓားဟာဒီဘယ်ကတိ
 21 ကော်ဖီသောက်လို့ရပါတယ်။ ။ ထွင်ဝန်းမှာအဆစ်တည်လျှံဟိုး
 22 ကျွန်တော်သားလေးအတွက်အရှုပ်တစ်ရှုပ်ဝယ်ချင်လို့ပါ။ ။ ? (. ၃ .) မှာဒီဖြည်းဘူးဟဲထွေးနာ
 23 ကျွန်တော်ရထားပြောင်းဖို့လိုအပ်လား။ ။ ၃ . ၇ % မှာဝါစီးသူယဉ်ဒိန်း (ဘူ

Fig. Generated text with syllable segmented monolingual corpus

Conclusion

- **An Emerging Frontier:** While LNNs are currently less mainstream than Transformers, they represent a significant shift toward more flexible, bio-inspired AI.
- **Continuous Real-Time Adaptation:** Unlike static models, LNNs adapt their parameters to changing data streams on the fly.
- **Optimized for the Edge:** Their compact footprint makes them ideal for low-power hardware, including drones, medical wearables, and autonomous vehicles.
- **Resilience to Noise:** LNNs excel at processing messy, real-world time-series data and remain stable even when encountering unfamiliar environments.
- **Transparent & Interpretable:** With fewer parameters and a structure rooted in differential equations, LNNs offer a clearer "window" into the model's logic than traditional black-box architectures.
- **Sustainable & Green AI:** By drastically reducing computational overhead, LNNs provide a path toward high-performance AI with a minimal carbon footprint.

Conclusion

- **Limited Application Scope:** LNNs are primarily designed for time-series and sequential data, making them unsuitable for static data tasks like image classification.
- **Training Complexity:** Training LNNs involves complex differential equations, which can be computationally intensive to compute during the training phase.
- **Vanishing Gradient Problem:** Like many Recurrent Neural Networks (RNNs), LNNs can still struggle with learning long-term dependencies over very long sequences.
- **Newer Technology:** As a relatively new architecture (developed by MIT in 2020), there is less literature, tooling, and industry experience compared to traditional models.
- **Difficult Hyperparameter Tuning:** Finding the right parameters for the liquid layer can be difficult due to its dynamic nature.

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Thank you!

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Talk

Ye Kyaw Thu (ရဲကျော်သူ)



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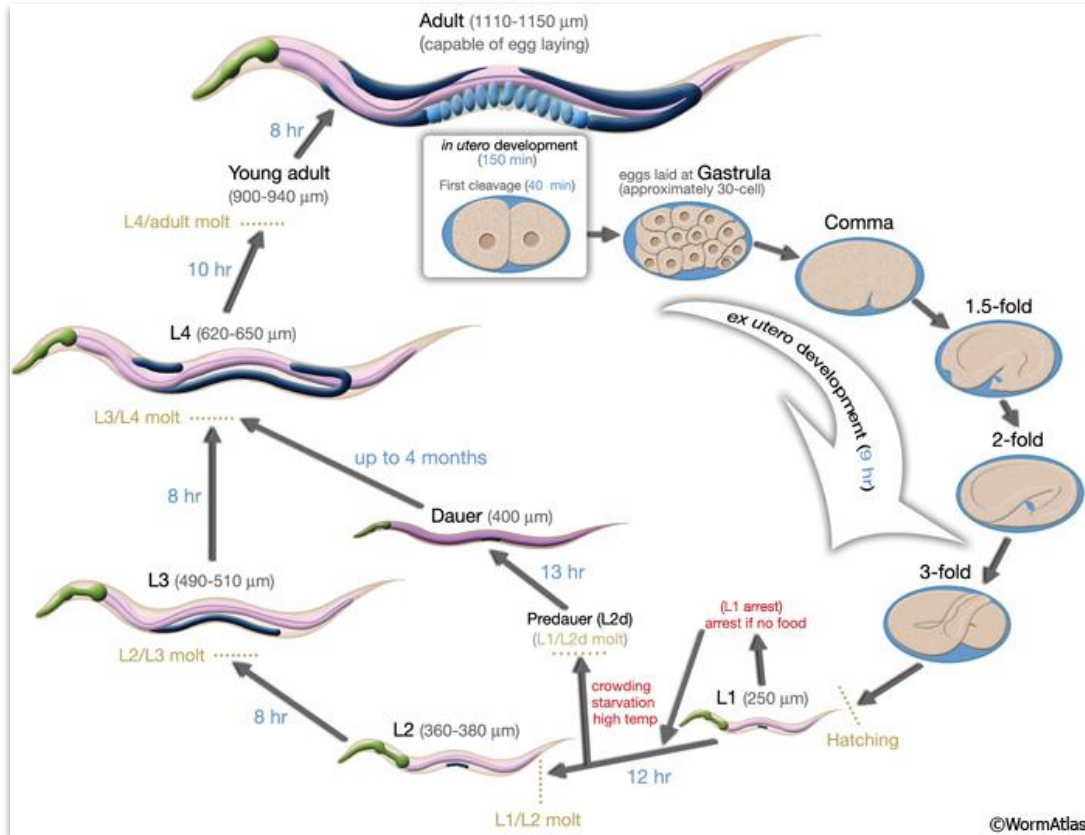
Lifespan of *C. elegans*

- Average lifespan: about 2–3 weeks
- Time to adulthood: ~3 days (from egg to adult)
- Reproductive period: days 3–10
- Old age / decline: after ~10 days

Lifespan of *C. elegans*

- *C. elegans* reproduces using hermaphrodites and males, with no females; hermaphrodites can self-fertilize or mate with males.
- A hermaphrodite has:
 - Male reproductive organs (produces sperm)
 - Female reproductive organs (produces eggs)
- A single hermaphrodite can:
 - Reproduce by itself (self-fertilization)
 - Or mate with a male
- Males make up only ~0.1–0.2% of wild populations, they arise due to chromosomal differences or errors, they exist mainly to increase genetic diversity

Lifespan of *C. elegans*



- *C. elegans* life cycle with dauer branch (20 C) from egg to reaching L4 (47 hours)

Sources:

<https://www.wormatlas.org/hermaphrodite/introduction/mainframe.htm>

<https://evandrofanglab.com/wp-content/uploads/2021/07/2021-07-08-how-to-work-in-the-lab-handing-c.-elegans-c.-elegans-other-details.pdf>

10 Reasons Why *C. elegans* is Essential to Science

- **Extreme Resilience:** They can survive catastrophic events and extreme environments, such as the disintegration of a space shuttle during re-entry, making them ideal for space-based research.
- **Genetic Pioneer:** *C. elegans* was the first animal in history to have its entire genome sequenced, providing a foundational template for understanding animal genetics.
- **Proven Scientific Value:** Research involving these worms has been so impactful that it has contributed to three Nobel Prizes.
- **Manageable Complexity:** While the human brain has 100 billion neurons, the worm has only about 300, making it a "simplified" model that is much easier for scientists to map and study in its entirety.
- **Advanced Behaviors:** Despite their small size, they exhibit complex behaviors—such as foraging, escaping threats, forming memories, and sensing magnetic fields—that mirror higher-order organisms.

10 Reasons Why C. elegans is Essential to Science

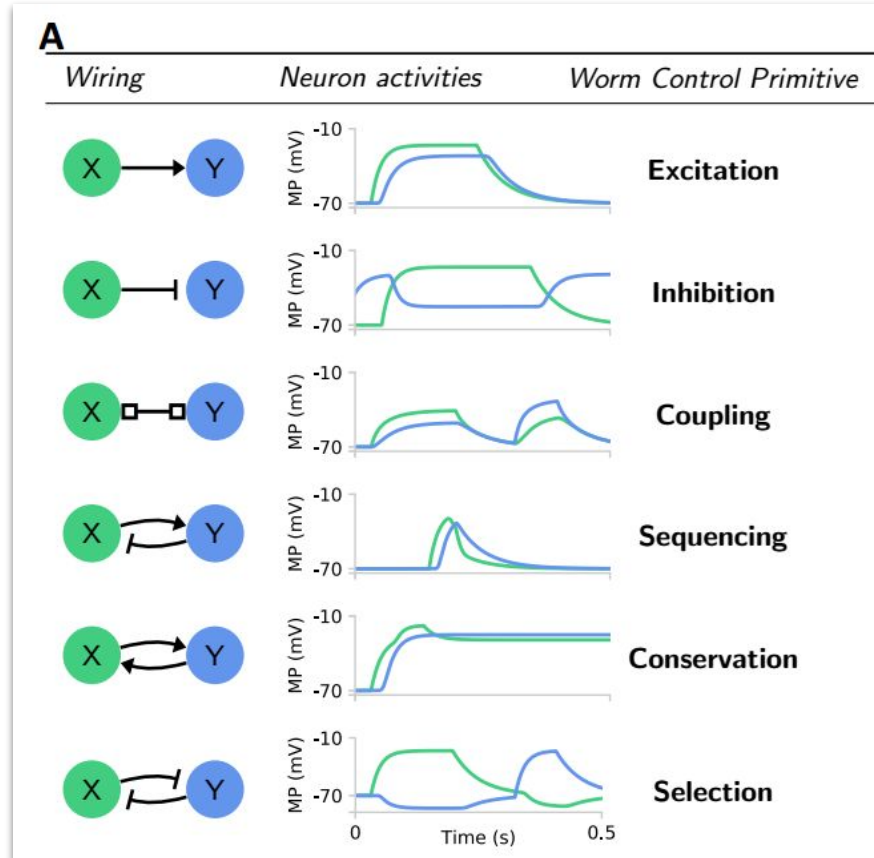
- The Complete Connectome: It is currently the only animal with a fully detailed map of its neural wiring (the "connectome"), a task that remains impossible for larger brains like those of mice or humans.
- Shared Genetics: Humans and C. elegans share significant genetic similarities, meaning discoveries made in the worm's brain can often be translated to help us understand the human brain.
- Chemical Network Mapping: They allow scientists to study "invisible" chemical networks (monoamines like dopamine and serotonin) which are crucial to understanding how brains regulate information and mood.
- Insights into Mental Health: By studying how monoamines coordinate in the worm, researchers can better understand the biological roots of human psychiatric conditions like depression and schizophrenia.
- Modeling Human Disorders: The worm exhibits behaviors that mirror human drug dependence and eating disorders, providing a unique opportunity to test theories on addiction and behavior in a controlled environment.

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Operators



- Design Operators. (A) Time series of the activity of DOs.
- From PhD thesis, Interpretable Recurrent Neural Networks in Continuous-time Control Environments, Vienna PhD School of Informatics by Ramin Hasani, Registration Number 01529745, 10 April 2020

Sources:

<https://repositum.tuwien.at/bitstream/20.500.12708/1068/2/Hasani%20Ramin%20-%202020%20-%20Interpretable%20recurrent%20neural%20networks%20in%20continuous-time...pdf>